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# Gamma hedging and rough paths

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**Abstract**
We apply rough-path theory to study the discrete-time gamma-hedging strategy. We show that if a trader knows that the market prices of a set of European options are given by a diffusive pricing model, then the discrete-time gamma-hedging strategy enables them to replicate other European options so long as the underlying pricing signal has finite $p$-variation for $p < 3$, with the error in the discrete-time replication strategy tending to zero as the length of the largest hedging interval tends to zero. This is a sure result and does not require that the underlying pricing signal has a quadratic variation corresponding to a probabilistic pricing model. We show how to generalise this result to exotic derivatives when the gamma is defined to be the Gubinelli derivative of the delta by deriving rough-path versions of the Clark–Ocone formula. We illustrate our theory by proving that if a stock price path has finite $p$-variation for $p < 3$ and if the implied volatility process for a European derivative on the stock (with a smooth, convex, nonlinear payoff and maturity $T$) has finite $q$-variation for $q < 2$ and $\frac{1}{p} + \frac{1}{q} > 1$, one can use the gamma-hedging strategy to replicate any European derivative with smooth payoff and maturity $T$. This is a sure result which holds without assuming any probabilistic model for the trajectory of the stock price path.

**Keywords** Gamma hedging · Rough path

**Mathematics Subject Classification** 91G80 · 60L20

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## 1 Introduction

With the discrete-time gamma-hedging strategy, a trader seeks to replicate a derivative product by trading in $n$ hedging instruments, such as the underlying and other

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derivatives. They do this by choosing a pricing model and then ensuring that at each time point, the total delta and total gamma of their hedging portfolio match that of the derivative they seek to replicate.

We prove that for many derivatives and pricing models, the profit and loss of this strategy converges to zero as the time between hedges tends to zero, under the assumption that the pricing model accurately prices the hedging instruments given a pricing signal derived from market data. This is a sure convergence result which holds so long as the pricing signal has sufficient regularity, as opposed to an almost sure result which only holds on a set of measure 1. There is no need for the pricing signal to be compatible with any probabilistic model, and we do not assume the existence of any physical probability measure for this signal.

We represent the *pricing signal* mathematically as a continuous function $S$ from $[0, T]$ to $U$, an open subset of $\mathbb{R}^d$, i.e., $S \in C^0([0, T]; U)$. This signal contains the information needed to determine the price of the derivatives that the trader uses for hedging. The signal may include the price of the underlying, but may also include information which is not directly traded. For example, the signal may encode the implied volatility of an option, or the current level of stochastic volatility. In addition, the payoffs of all derivatives that we consider must be functions of $S$.

Let us give a precise statement in the case of European options. For simplicity, we assume that there is a risk-free asset with deterministic interest rate 0; this amounts to making an appropriate choice of numeraire. **We assume throughout** that $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{Q})$ is a filtered probability space with $\mathbb{F} = (\mathcal{F}_t)_{t\in[0,T]}$ generated by a $d$-dimensional Brownian motion $(W_t)$ and satisfying the usual conditions. We write $\mathbb{E}[X]$ for the expectation of a random variable $X$ under $\mathbb{Q}$ and set $\mathbb{E}_t[X] := \mathbb{E}[X|\mathcal{F}_t]$. We write $s_\alpha$ $(\alpha = 1, \ldots, d)$ for the individual coordinates of $s \in U$ and $\frac{\partial}{\partial s_\alpha}$ to denote partial derivatives with respect to these coordinates for $U$. When we say that a function is smooth with no more precise qualification, we mean $C^\infty$-smoothness.

**Theorem 1.1** *Let $S \in C^0([0, T]; U)$ be a pricing signal. Let $f^i : U \to \mathbb{R}$ $(i = 0, \ldots, n)$ be smooth payoff functions. Let $V^i : U \times [0, T] \to \mathbb{R}$ be defined by*

$$V^i(s, u) := \mathbb{E}[f^i(\tilde{S}_T^{s,u})], \tag{1.1}$$

*where $(\tilde{S}_t^{s,u})_{t\in[u,T]}$ is the solution of*

$$d\tilde{S}_t = \mu(\tilde{S}_t, t)\, dt + \sigma(\tilde{S}_t, t)\, dW_t \tag{1.2}$$

*with initial condition $\tilde{S}_u = s$. Here $\mu$ and $\sigma$ are smooth functions defined on $U \times [0, T]$ and taking values in $\mathbb{R}^d$ and the space of $d \times d$ matrices, respectively. We assume that the SDE almost surely has a strong solution satisfying $\tilde{S}_t^{s,u} \in U$ for all $t \in [u, T]$ and that the $V^i$ are smooth. We suppose that for $1 \le i \le n$, the option with payoff $f^i(S_T)$ can be continuously bought and sold at the price $V^i(S_t, t)$. Suppose*

*that quantities $q_t^i$ can be found satisfying*

$$\sum_{i=1}^{n} q_t^i \frac{\partial}{\partial s_\alpha} V^i(S_t, t) = \frac{\partial}{\partial s_\alpha} V^0(S_t, t), \tag{1.3}$$

$$\sum_{i=1}^{n} q_t^i \frac{\partial^2}{\partial s_\beta \partial s_\alpha} V^i(S_t, t) = \frac{\partial^2}{\partial s_\beta \partial s_\alpha} V^0(S_t, t) \tag{1.4}$$

*for all indices $1 \le \alpha, \beta \le d$ and $t \in [0, T]$. The* discrete-time gamma-hedging strategy *for a partition of $[0, T]$ is to hold a quantity $q_{t_k}^i$ units of option $i$ for each time interval $[t_k, t_{k+1})$ in the partition, and to place any remaining wealth in the risk-free account. At time $T$, the portfolio is liquidated.*

*Suppose that the initial wealth of the trader is $V^0(S_0, 0)$, the $q_t^i$ are uniformly bounded and $S$ has finite $p$-variation for some $p < 3$. Let $(\pi_j)_{j\in\mathbb{N}}$ be any sequence of partitions of the interval $[0, T]$ with mesh size $|\pi_j|$ tending to 0 as $j \to \infty$. Then the final value of the discrete-time gamma-hedging strategy for the partition $\pi_j$,*

$$V^0(S_0, 0) + \sum_{i=1}^{n} \sum_{[t_k, t_{k+1}]\in\pi_j} q_{t_k}^i \big(V^i(S_{t_{k+1}}, t_{k+1}) - V^i(S_{t_k}, t_k)\big),$$

*tends to $f^0(S_T)$ as $j \to \infty$.*

Our assumption that the derivative prices are given by $V^i(S_t, t)$ explains why we call $S$ a pricing signal. Notice that the right-hand side of (1.1) is not a random variable and could by the Feynman–Kac theorem be expressed by using PDE notation with no mention of a stochastic process. We allow a drift term in (1.2) to allow the possibility that some components of the signal may not represent traded assets.

Our requirement that the $V^i$ are smooth can be ensured for example by requiring that $\sigma$ has bounded derivatives of all orders and satisfies the Hörmander condition; this requires that the vector space spanned by repeated Lie brackets of its column vectors at each point and by the drift vector is equal to $\mathbb{R}^d$ (see Nualart [37, Theorem 2.3.3]). If the signal is a typical stock price process, one needs to apply a logarithmic transformation to the SDE before applying a general-purpose theorem of this sort.

For any fixed set of $n$ options, it may not always be possible to find quantities which solve (1.3) and (1.4); so this is an assumption of our theorem. In practice, a trader is likely to be using exchange-traded derivatives to hedge an over-the-counter derivative. There are normally a large number of exchange-traded derivatives available; so this is not a very restrictive assumption.

Theorem 1.1 can be interpreted as a theoretical model for the success of the market practice of choosing a pricing model by calibrating to market options prices and then pursuing the gamma-hedging strategy without ever formulating a physical-measure model for the trajectories of the underlying. In classical delta hedging, one assumes the underlying follows a physical measure that is equivalent to the pricing measure and obtains at best almost sure results. Neither of these restrictions apply to gamma

hedging. We illustrate this in detail with the example of calibrating the Black–Scholes model in Sect. 4.1.

If we assume that some parts of the signal are more regular, then it is not necessary to match all the second-derivative terms. We illustrate this by proving that if the stock price process is sufficiently regular, as is the implied volatility process of a European derivative with maturity $T$ and smooth payoff $f(S_T)$ satisfying $f'' > 0$, one can replicate with certainty any European derivative with smooth payoff and maturity $T$ using just the stock and the derivative with payoff $f$. A precise statement is given in Theorem 4.3.

Although the statement of Theorem 1.1 does not mention rough-path theory, the latter is the basic tool that we use to prove our results. In rough-path theory, if one wishes to integrate some integrand against a signal, then the integrand must be similar to the signal in the sense that it must possess a so-called Gubinelli derivative. The rough-path integral then consists of two terms: a classical term and a term involving the Gubinelli derivative. The classical theory of delta hedging proceeds by using the martingale representation theorem to show that the payoff of a derivative can be written as an integral of the delta against the changes in the underlying and then matching the delta of the hedging portfolio to that of the instrument to be replicated. When using rough-path theory, we must also match the terms arising from the Gubinelli derivatives. In the case of European options, we shall see that the Gubinelli derivative of the delta is equal to the gamma of the option, and we use this to define the gamma for path-dependent derivatives.

Rough-path theory has been used in model-free finance before. This paper builds on Armstrong et al. [4] which shows how an enhanced version of the classical delta hedge is required for pathwise replication for European options, proposing a theoretical strategy of enhancing delta hedging with trades in a hypothetical form of volatility swap. The mathematics behind our approach has its origins in Föllmer's [18] paper "Calcul d'Itô sans probabilités" which defines Itô integration pathwise and proves a pathwise Itô formula so long as the integrator admits a finite quadratic variation. Bick and Willinger [8] have shown this can be used to obtain a pathwise theory of delta hedging subject to some restrictions on the grid of points used when discretising the delta-hedging strategy. Perkowski and Prömel [39] studied two model-free approaches to stochastic integration, one using the outer Vovk measure and the other rough-path integration, and Allan et al. [2] extend this to include signals with jumps. Numerical results have been obtained in the model-free pricing and hedging of exotic options by Lyons et al. [34] by using signature payoffs.

Our results are also related to functional Itô calculus developed by Cont and Fournié [11] and Cont [10, Chaps. 4–6], which extends Itô calculus to path-dependent functionals of stochastic processes using the derivatives of Dupire [15]. The PhD thesis of Riga [41, Chaps. 3 and 4] analyses continuous-time trading strategies pathwise; see also Ananova [3] which relates rough differential equations with path-dependent coefficients to the weak derivative of functional Itô calculus. We do not consider functional Itô calculus in this paper, but note that it is possible to relate the gamma as defined in this paper as a Gubinelli derivative with the natural definitions in terms of Dupire's derivatives; details can be found in Ionescu [31, Proposition 13.2.18].

The topic of gamma hedging has been studied previously. In practice, delta hedging in discrete time produces much larger hedging errors than gamma hedging does.

Thus it is common for traders to augment their strategies with gamma (or vega) terms to reduce this hedging error and reduce transaction costs. For the case of delta hedging smooth European payoffs, Heston and Zhou [28] show that the hedging error is inversely proportional to the number $N$ of rebalancing times. For gamma hedging, Gobet and Makhlouf [25] give non-sharp lower bounds on the convergence rates for both equidistant and non-equidistant trading grids. They show a convergence rate of $N^{\frac{1}{2}}$ even for non-smooth payoffs. Brodén and Wiktorsson [9] partially improve on this result by proving that gamma hedging with European options on an equidistant time grid in the Black–Scholes model has an order of convergence of $N^{\frac{3}{4}}$. For the efficient computation of the gamma of options, Fournié et al. [19, 20] use Malliavin calculus to compute the gamma of "exotic European options" in the Black–Scholes model. Gobet and Kohatsu-Higa [24] also use Malliavin calculus to compute the Greeks of barrier and lookback options. Dengler and Jarrow [13] develop a discrete-time binomial option pricing model with random time steps. They show that under this model, delta hedging approximates gamma hedging in the classical binomial model with fixed time steps, which in the continuous-time limit becomes the Black–Scholes model.

Theorem 1.1 shows that the gamma-hedging strategy is extremely robust to a misspecification of the probability model for the underlying, though it introduces a new dependency on the pricing model that determines how market prices for the hedging options react to changes in the underlying. Our result can therefore be contrasted with the literature on the effect of hedging under a misspecified volatility. El Karoui et al. [16] showed that a hedging strategy computed under the assumption that the misspecified volatility dominates the true volatility provides a one-sided hedge for both European and American options. Analogous results are also shown when the true volatility dominates the misspecified volatility. Ahmad and Wilmott [1] derive the formula for profits from delta hedging European options under a three-part structure: they explicitly differentiate between a true governing volatility of the underlying asset, inaccessible to the trader, an implied volatility characterised by option prices in the market, and a hedge volatility reflecting the personal views of the trader. A thorough exposition of this result, known as the fundamental theorem of derivatives trading, can be found in Ellersgaard et al. [17].

More generally, there is an extensive literature on model-independent finance. Examples include Lyons [35] which describes optimal and risk-free hedging strategies when the volatility is unknown, but is assumed to lie in some convex region depending on market prices. Avellanada et al. [5] describe pricing and hedging methods when the volatility is assumed to lie in some interval $(\sigma_{\min}, \sigma_{\max})$; see also Hobson [30] and the subsequent development of the Skohorod embedding problem (SEP) to develop robust, model-independent prices and hedges for exotic options described in [29] and the survey article by Obłój [38]. It is related to the problem of martingale optimal transport (MOT) introduced by Beiglböck et al. [7]. In Backhoff-Veraguas et al. [6], hedges are shown to be robust under changes in the risk-neutral measure if one uses the adapted Wasserstein distance to measure the proximity of the model used for hedging to the model used to generate the path. The superreplication approach and the MOT problem are shown to be dual problems by Dolinsky and Soner [14].

The first contribution of the current paper is to consider the gamma-hedging strategy using the techniques of rough-path theory. In particular, we show that the trading

strategy in volatility swaps proposed in Armstrong et al. [4] can be replaced with a gamma-hedging strategy in conventional derivatives.

The second contribution of this paper is to show how Theorem 1.1 can be generalised to path-dependent derivatives. This requires identifying the Gubinelli derivative of the delta of an option (if it exists). Whereas [4] applies PDE techniques to the analysis of European options, we show how to identify the gamma for large classes of derivatives using probabilistic techniques and how it can be related to the Malliavin derivative of the delta. We show that barrier options do not in general admit such a gamma due to discontinuities of the delta when the barrier is hit, but we also show how barrier options can be approximated by instruments with smoother payoffs which do have a gamma. Indeed, we show that all bounded derivatives with a payoff that is continuous in the sup-norm on the space of paths can be surely superhedged for a price arbitrarily close to the risk-neutral price.

We remark that if one is content with probabilistic results when the underlying is a semimartingale, then Theorem 1.1 can be proved using classical stochastic calculus techniques; but we only consider the rough-path approach in this paper. One reason for preferring the rough-path approach is that it applies even if the signal does not arise from a probability model that is independent of the trader's actions; indeed, the signal could be chosen by an adversary who already knows our strategy.

Section 2 gives a brief outline of the notation we use from rough-path theory. Section 3 proves a general result showing the sure convergence of the gamma-hedging strategy under the assumption that rough-path integral formulae analogous to those in the martingale representation theorem hold. Section 4 applies this to European options, giving a proof of Theorem 1.1. Section 5 extends the analysis to path-dependent derivatives. Our proof strategy is to use the Clark–Ocone formula to obtain almost sure results allowing us to write the payoff as an integral and to identify the gamma. We then use the relationship between the Itô integral and the rough integral, together with the continuity of the rough integral, to obtain sure results. Section 5.1 describes this general theory; Sect. 5.2 applies this to an $L^2$-dense family of payoffs we call simple payoffs, to signature payoffs and to barrier options. Section 5.3 then demonstrates how one can superhedge derivatives surely.

## 2 Rough-path theory notation

In this section, we describe the notation we use for rough-path theory. Unless stated otherwise, all definitions below are taken from Friz and Hairer [21, Chaps. 2 and 4] and adjusted to appropriate $p$-variation regularity versions using Friz and Victoir [22, Chaps. 5 and 6]. We use the $p$-variation formulation as it makes our theory of controllable options easier to prove for one-touch options: the running maximum process is increasing, so it has finite 1-variation and hence finite $p$-variation for $p \geq 1$.

### 2.1 Rough paths

Given a metric space $(E, d)$, a function $f : [0, T] \to E$ and $p > 1$, we write $\|f\|_{p\text{-var};[0,T]}$ for the $p$-variation of $f$ (see for example [22, Definition 5.1]) if it

exists. We write $C^{p\text{-var}}([0,T];E)$ for the set of all continuous functions from $[0,T]$ to $E$ with finite $p$-variation.

Given a Banach space $\mathcal{V}$ and an interval $[s,t] \subseteq \mathbb{R}$, we define a $p$-variation norm for functions of two variables $\mathbb{F}_{(\cdot,\cdot)} : [s,t] \times [s,t] \to \mathcal{V}$ by

$$\|\mathbb{F}\|_{p\text{-var};[s,t]} := \left( \sup_{\pi \in \mathcal{P}[s,t]} \sum_{[t_k,t_{k+1}] \in \pi} |\mathbb{F}_{t_k,t_{k+1}}|^p \right)^{\frac{1}{p}},$$

where $\mathcal{P}[s,t]$ is the set of partitions of $[s,t]$. We abbreviate $\|\mathbb{F}\|_{p\text{-var};[0,T]}$ to $\|\mathbb{F}\|_{p\text{-var}}$. We write $C^{p\text{-var}}([0,T]^2;\mathcal{V})$ for the space of continuous functions with finite $p$-variation. We write $|\pi|$ for the mesh size of a partition, that is, the length of the largest sub-interval. Given a path $X : [0,T] \to \mathcal{V}$, we write $X_{s,t} := X_t - X_s$.

**Definition 2.1** For $p \in (2,3)$, we write $\mathcal{C}^{p\text{-var}}([0,T];\mathcal{V})$ for the *space of $p$-variation rough paths* over $\mathcal{V}$ as those pairs

$$\mathbf{X} := (X, \mathbb{X}) \in C^{p\text{-var}}([0,T];\mathcal{V}) \times C^{\frac{p}{2}\text{-var}}([0,T]^2; \mathcal{V} \otimes \mathcal{V})$$

satisfying

1) $\|X\|_{p\text{-var}} + \|\mathbb{X}\|_{\frac{p}{2}\text{-var}} < \infty$;

2) *Chen's relation*: for all triples $(s,u,t) \in [0,T]^3$, we have

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}.$$

Given $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^{p\text{-var}}([0,T];\mathcal{V})$, we call the $\mathcal{V}$-valued path $X$ its *trace* and $\mathbb{X}$ its *lift*. The norm of the rough path is defined to be

$$|||\mathbf{X}|||_{p\text{-var}} := \|X\|_{p\text{-var}} + \sqrt{\|\mathbb{X}\|_{\frac{p}{2}\text{-var}}},$$

with the square-root ensuring that this scales homogeneously. For $\mathbf{X} = (X, \mathbb{X})$ and $\mathbf{Y} = (Y, \mathbb{Y})$ in $\mathcal{C}^{p\text{-var}}([0,T];\mathcal{V})$, the (inhomogeneous) $p$-variation rough-path metric is defined to be

$$d_{p\text{-var}}(\mathbf{X},\mathbf{Y}) := \|X - Y\|_{p\text{-var}} + \|\mathbb{X} - \mathbb{Y}\|_{\frac{p}{2}\text{-var}}.$$

We also recall a few technical definitions (see [22, Definitions 1.6, 1,7]). Let $\Delta_T$ be the simplex $\{(s,t) \in [0,T] \times [0,T] : 0 \le s \le t \le T\}$. A map $w : \Delta_T \to [0,\infty)$ is said to be *superadditive* if for all $s \le u \le t$ in $[0,T]$, we have

$$w(s,u) + w(u,t) \le w(s,t).$$

If $w$ in addition is continuous, we say that $w$ is a *control function* on $[0,T]$. Given an element $X \in C^{p\text{-var}}([0,T];\mathcal{V})$, we write $w_{X,p}(s,t) := \|X\|^p_{p\text{-var};[s,t]}$ for the control function defined by the given power of $p$-variation over the interval $[s,t]$. This is called the *$p$-variation control*. For $\mathbb{X} \in C^{p\text{-var}}([0,T]^2;\mathcal{V})$, we define the $p$-variation control by $w_{\mathbb{X},p} := \|\mathbb{X}\|^p_{p\text{-var};[s,t]}$.

**Lemma 2.2** *The sum of two control functions is a control function. If $w_1$ and $w_2$ are control functions and if $\alpha, \beta > 0$ with $\frac{1}{\alpha} + \frac{1}{\beta} > 1$, then $w_1^\alpha w_2^\beta$ is a control function.*

***Proof*** See [22, Exercise 1.10]. □

Given a finite partition $\pi \in \mathcal{P}[0, T]$ and a control $w$, the *modulus of continuity* of $w$ on a scale smaller than or equal to the mesh size $|\pi|$ is defined as

$$\operatorname{osc}(w, |\pi|) := \sup\{w(s, t) : |t - s| \leq |\pi|\}.$$

Control functions are useful in integration theory as for any $\epsilon > 0$, we have that

$$\begin{aligned}\sum_{[s,t]\in\pi} w(s,t)^{1+\epsilon} &\leq \Big(\sup_{[s,t]\in\pi} w(s,t)^\epsilon\Big)\Big(\sum_{[s,t]\in\pi} w(s,t)\Big)\\ &\leq \operatorname{osc}(w^\epsilon, |\pi|)w(0,T)\\ &\longrightarrow 0 \qquad \text{as } |\pi| \to 0. \end{aligned} \tag{2.1}$$

### 2.2 Controlled rough paths

Let $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^{p\text{-var}}([0, T]; \mathcal{V})$ be given. Let $\hat{\mathcal{V}}$ denote a Banach space, possibly different from $\mathcal{V}$. Let $\mathcal{L}(\mathcal{V}; \hat{\mathcal{V}})$ denote the space of continuous linear maps from $\mathcal{V}$ to $\hat{\mathcal{V}}$.

A pair $(Y, Y') \in C^{p\text{-var}}([0, T]; \hat{\mathcal{V}}) \times C^{p\text{-var}}([0, T]; \mathcal{L}(\mathcal{V}; \hat{\mathcal{V}}))$ is said to be an *$X$-controlled path of $p$-variation regularity* if the remainder $R^Y_{s,t} := Y_{s,t} - Y'_s X_{s,t}$ satisfies $R^Y \in C^{\frac{p}{2}}([0, T]^2; \mathcal{V})$. We write $\mathcal{D}_X^{p\text{-var}}([0, T]; \hat{\mathcal{V}})$ for the set of all $X$-controlled *paths of $p$-variation regularity* taking values in $\hat{\mathcal{V}}$. If the domain of $Y$ is clear from the context, we abbreviate this to $\mathcal{D}_X^{p\text{-var}}$. If $(Y, Y') \in \mathcal{D}_X^{p\text{-var}}$, we call $Y'$ a *Gubinelli derivative* of $Y$ with respect to $X$. The space $\mathcal{D}_X^{p\text{-var}}$ is a Banach space with norm

$$\begin{aligned}\|(Y, Y')\|_{\mathcal{D}_X^{p\text{-var}}} &:= |Y_0| + |Y'_0| + \|Y'\|_{p\text{-var}} + \|R^Y\|_{\frac{p}{2}\text{-var}}\\ &=: |Y_0| + |Y'_0| + \|Y, Y'\|_{X,p\text{-var}}\end{aligned}$$

If $(\tilde{X}_t)_{t\in[0,T]}$ is a diffusion process defined on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, then for $\omega \in \Omega$, $\tilde{X}(\omega)$ defines a path. For $p > 2$, $\tilde{X}(\omega)$ almost surely has finite $p$-variation and the Gubinelli derivative of $\tilde{X}(\omega)$ is almost surely unique (see the discussion of truly-rough paths in [21, Chap. 6]). Therefore in our applications to finance, we are often able to talk unambiguously of the Gubinelli derivative.

### 2.3 The rough integral

In rough-path theory, rough paths play the role of integrators and controlled rough paths play the role of integrands. Together they allow us to define the rough integral.

If we wish to integrate against $\mathbf{X} \in \mathcal{C}^{p\text{-var}}([0,T];\mathcal{V})$, then the integrand at each time should be a linear map from $\mathcal{V}$ to a target space $\hat{\mathcal{V}}$. This leads us to consider $X$-controlled paths with values in $\mathcal{L}(\mathcal{V};\hat{\mathcal{V}})$. The next result which allows us to define the rough integral is due to Gubinelli [26, Theorem 1].

**Theorem 2.3** *Given an $X$-controlled path*

$$(Y, Y') \in \mathcal{D}_X^{p\text{-var}}\big([0,T];\mathcal{L}(\mathcal{V};\hat{\mathcal{V}})\big),$$

*the limit*

$$\int_0^T Y\, d\mathbf{X} := \lim_{|\pi|\to 0} \sum_{[s,t]\in\pi} (Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t}) \in \hat{\mathcal{V}} \tag{2.2}$$

*exists and satisfies the estimate*

$$\left\| \int_s^t Y\, d\mathbf{X} - Y_s X_{s,t} - Y'_s \mathbb{X}_{s,t} \right\| \le C(p)\omega(s,t)^{\frac{3}{p}},$$

*where the constant $C(p)$ is global and only depends on $p$, and $w(s,t)$ is the control function*

$$w(s,t) = \|R^Y\|^{\frac{p}{3}}_{\frac{p}{2}\text{-var};[s,t]} \|X\|^{\frac{p}{3}}_{p\text{-var};[s,t]} + \|Y'\|^{\frac{p}{3}}_{p\text{-var};[s,t]} \|\mathbb{X}\|^{\frac{p}{3}}_{\frac{p}{2}\text{-var};[s,t]}.$$

*Moreover, the map*

$$(Y, Y') \mapsto (Z, Z') := \left( \int_0^{\cdot} Y\, d\mathbf{X}, Y \right)$$

*from $\mathcal{D}_X^{p\text{-var}}([0,T];\mathcal{L}(\mathcal{V};\hat{\mathcal{V}}))$ to $\mathcal{D}_X^{p\text{-var}}([0,T];\hat{\mathcal{V}})$ is continuous.*

When we wish to highlight the role of the Gubinelli derivative, we use the notation $((Y,Y')\cdot(X,\mathbb{X})) := \int Y\, d\mathbf{X}$.

The rough integral allows one to define rough differential equations and prove existence and uniqueness of solutions under appropriate assumptions on the coefficients; see [21, Chap. 8]. The solution map for rough differential equations is continuous in the driving noise in the rough-path topology.

## 3 Gamma hedging

In this section, we state and prove our main result, which is to define the gamma-hedging strategy and to estimate its error. For simplicity, we consider a financial market with risk-free rate 0.

**Theorem 3.1** *Let $(S,\mathbb{S}) \in \mathcal{C}^{p\text{-var}}([0,T];\mathbb{R}^d)$ be a rough path whose trace represents a pricing signal. Assume the market contains $n$ financial instruments that we use for*

*hedging and that the price $V^i_t$ of instrument $i$ ($i = 1, \dots, n$) at each time $t$ satisfies the rough-integral formula*

$$V^i_t = V^i_0 - \int_0^t \Delta^i_u m_u \, du + \big((\Delta^i, \Gamma^i) \cdot (S, \mathbb{S})\big)_{0,t}, \tag{3.1}$$

*where $m \in C^0([0,T]; \mathbb{R}^d)$, the first integral is a Riemann integral and*

$$(\Delta^i, \Gamma^i) \in \mathcal{D}_S^{p\text{-var}}\big([0,T]; \mathcal{L}(\mathbb{R}^d, \mathbb{R})\big).$$

*Suppose that $V^0_t$ also satisfies (3.1) and that we can find bounded quantities $q^i_t$ such that for all $t \in [0,T]$,*

$$\begin{cases} \sum_{i=1}^n q^i_t \Delta^i_t = \Delta^0_t, \\ \sum_{i=1}^n q^i_t \Gamma^i_t = \Gamma^0_t. \end{cases} \tag{3.2}$$

*Let $\pi \in \mathcal{P}[0,T]$ be a grid of time points. Suppose a trader has initial wealth $V^0_0$ and ensures that they hold $q^i_{t_k}$ units of each hedging instrument $i$ at each time $t \in [t_k, t_{k+1})$ for each $[t_k, t_{k+1}] \in \pi$. They place the rest of their wealth in the risk-free account. At time $T$, the portfolio is liquidated. This is the discrete-time gamma-hedging strategy. If $\Pi^\pi_t$ denotes the wealth of the trader at time $t$, we have $|\Pi^\pi_T - V^0_T| \to 0$ as $|\pi| \to 0$.*

***Proof*** We allow $C$ to vary line by line in our proof in the usual way. We also set $q^0 \equiv -1$. Let $[s,t] \in \pi$ be an interval of our discrete trading grid. The increment $\Pi^\pi_{s,t}$ of our wealth from $s$ to $t$ satisfies

$$\Pi^\pi_{s,t} = \sum_{i=1}^n q^i_s V^i_{s,t}.$$

Hence

$$\begin{aligned} |\Pi^\pi_T - V^0_T| &= |\Pi^\pi_{0,T} - (V^0_T - V^0_0)| \\ &= \left| \sum_{[s,t]\in\pi} \Pi^\pi_{s,t} + \int_0^T \Delta^0_u m_u \, du - \int_0^T \Delta^0 \, d\mathbf{S} \right| \\ &= \left| \sum_{[s,t]\in\pi} \left( \sum_{i=1}^n q^i_s V^i_{s,t} + \int_s^t \Delta^0_u m_u \, du - \int_s^t \Delta^0 \, d\mathbf{S} \right) \right| \\ &= \left| \sum_{[s,t]\in\pi} \sum_{i=0}^n q^i_s \left( - \int_s^t \Delta^i_u m_u \, du + \int_s^t \Delta^i \, d\mathbf{S} \right) \right|, \end{aligned} \tag{3.3}$$

where we have used (3.1) in the last line. We first bound the rough-integral term in this expression. Using (3.2) and the triangle inequality, we get

$$\begin{aligned}\left|\sum_{i=0}^{n} q_s^i \int_s^t \Delta^i \, d\mathbf{S}\right| &= \left|\sum_{i=0}^{n} q_s^i \int_s^t \Delta^i \, d\mathbf{S} - \sum_{i=0}^{n} q_s^i (\Delta_s^i S_{s,t} + \Gamma_s^i \mathbb{S}_{s,t})\right| \\ &\leq \sum_{i=0}^{n} |q_s^i| \left| \int_s^t \Delta^i \, d\mathbf{S} - (\Delta_s^i S_{s,t} + \Gamma_s^i \mathbb{S}_{s,t}) \right| \\ &\leq C\Big(\max_{0\leq i\leq n} \|q^i\|_\infty\Big) \sum_{i=0}^{n} w_i(s,t)^{\frac{3}{p}},\end{aligned}$$

where $w_i$ denotes the control function defined in Theorem 2.3 for the integral of $\Delta^i$. Since $p < 3$ and $w_i$ is a control function, we may use the estimate (2.1) to see that the size of the rough-integral terms in (3.3) tends to 0 as $|\pi| \to 0$.

We now bound the Riemann integral in (3.3) in a similar fashion. Using (3.2) as before, we find

$$\begin{aligned}\left|\sum_{i=0}^{n} q_s^i \int_s^t \Delta_u^i m_u \, du\right| &\leq \left|\sum_{i=0}^{n} q_s^i \int_s^t \Delta_u^i m_u \, du - \sum_{i=0}^{n} q_s^i (\Delta_s^i m_s)(t-s)\right| \\ &\leq \sum_{i=0}^{n} |q_s^i| \left| \int_s^t \Delta_u^i m_u \, du - (\Delta_s^i m_s)(t-s) \right| \\ &\leq C \sum_{i=0}^{n} |q_s^i| \operatorname{osc}(\Delta^i m, |\pi|)|t-s|.\end{aligned}$$

We have used bounds on the Riemann integral to obtain the final line. Hence

$$\begin{aligned}\left|\sum_{[s,t]\in\pi} \sum_{i=0}^{n} q_s^i \int_s^t \Delta_u^i m_u \, du\right| &\leq CT \max_{0\leq i\leq n} \|q^i\|_\infty \operatorname{osc}(\Delta^i m, |\pi|) \\ &\longrightarrow 0 \qquad \text{as } |\pi| \to 0.\end{aligned}$$

This completes the proof. □

***Remark 3.2*** Note that we can weaken the second equation in (3.2) by only requiring that there exists a lift $\mathbb{S}_{s,t}$ such that

$$\sum_{i=1}^{n} q_s^i \Gamma_s^i \, \mathbb{S}_{s,t} = \Gamma_s^0 \, \mathbb{S}_{s,t}, \qquad 0 \leq s < t \leq T.$$

This may make it possible to reduce the number of required hedging instruments.

According to the terminology of Theorem 3.1, the gamma-hedging strategy is only defined when (3.1) holds for all hedging assets, and this equation can be taken as

defining the meaning of $\Delta$ and $\Gamma$. We shall see how these can be related to the familiar Greeks of European options in Sect. 4. This will justify our notation and choice of terminology. We consider path-dependent derivatives in Sect. 5.

In Theorem 3.1, the pricing signal $S$ need not be traded. However, one specific application we have in mind is when some component $S^\alpha$ of $S$ represents the price path of an underlying asset and one of the derivatives also represents this underlying and thus has price $V^i = s_\alpha$. In these circumstances, if $e_\alpha \in \mathcal{L}(\mathbb{R}^d; \mathbb{R})$ is the coordinate functional with $e_\alpha(s) = s_\alpha$, the constant path $\mathbf{e}_\alpha := (e_\alpha, 0)$ is an $S$-controlled rough path and satisfies (3.1) if we take $\Delta^i = e_\alpha$, $\Gamma^i = 0$ and if $m \in \ker e_\alpha$.

## 4 European options

In this section, we consider the case when all assets of Theorem 3.1 represent European derivatives on $S$ with terminal payoffs $f^i(S_T)$. In this case, we are able to find conditions when (3.1) holds by using the results of Armstrong et al. [4] which we now summarise.

Let $\sigma : U \times [0,T] \to \mathbb{R}^{d\times d}$ and $\mu : U \times [0,T] \to \mathbb{R}^d$ be smooth functions. Suppose that the prices of the hedging instruments at time $t$ are given by $V^i(S_t, t)$, where the functions $V^i : U \times [0,T] \to \mathbb{R}$ are smooth and satisfy the Feynman–Kac PDE, for $(s,t) \in U \times [0,T]$,

$$0 = \frac{\partial}{\partial t} V^i(s,t) + \sum_{\alpha=1}^{d} \mu_\alpha(s,t) \frac{\partial}{\partial s_\alpha} V^i(s,t) + \frac{1}{2} \sum_{1\le\alpha,\beta\le d} \left(\sigma(s,t)\sigma(s,t)^\top\right)_{\alpha,\beta} \frac{\partial^2}{\partial s_\alpha \partial s_\beta} V^i(s,t) \tag{4.1}$$

with boundary condition $V^i(s,T) = f^i(s)$ for all $s \in U$. Suppose also that $(S, \mathbb{S})$ is a rough path. We can combine this with the smooth path $t$ with lift 0 to obtain a rough path $(X, \mathbb{X})$ with trace $X = (S, 0)$. Letting $\nabla$ denote the gradient operator, each $(\nabla V^i(X), \nabla^2 V^i(X))$ is then an $X$-controlled rough path. Let us write

$$\left.\frac{\partial V^i}{\partial s_\alpha}\right|_u := \frac{\partial V^i}{\partial s_\alpha}(S_u, u)$$

for the partial derivatives evaluated along the path $S$, with a similar notation for higher-order partial derivatives and the operator $\nabla$. Applying the rough-path Itô lemma (see Friz and Hairer [21, Proposition 5.8]) to $\mathbf{X}$, we find that

$$V^i(S_t,t) = V^i(S_0,0) + \int_0^t \nabla V^i|_u \, dS_u + \int_0^t \left.\frac{\partial V^i}{\partial t}\right|_u du + \int_0^t \frac{1}{2}\nabla^2 V^i|_u \, d[\mathbf{S}]_u, \tag{4.2}$$

where the *rough bracket* $[\mathbf{S}]$ is defined by

$$[\mathbf{S}]_t = S_{0,t} \otimes S_{0,t} - 2\,\mathrm{Sym}(\mathbb{S}_{0,t}),$$

with Sym denoting tensor symmetrisation. In (4.2), the first integral is a rough-path integral, the second a Riemann integral and the third a Young integral.

If we can assume that

$$[\mathbf{S}]_t = \int_0^t \sigma(S_u, u)\sigma(S_u, u)^\top \, du, \tag{4.3}$$

then (4.1) and (4.2) will together imply (3.1) when $m_t = \mu(S_t, t)$, $\Delta^i = \nabla V^i$ and $\Gamma^i = \nabla^2 V^i$. It is not obvious that given a path $S$ of finite $p$-variation, we can then find a lift $\mathbb{S}$ such that (4.3) holds. However, we note that in the rough-path Itô formula (4.2), the integrand in the final integral is the symmetric tensor $\nabla^2 V^i|_u$. As a result, we do not need to identify a full lift for $S$, but only the symmetric part of the lift. This motivates the definition of reduced rough paths; see [21, Definition 5.3].

**Definition 4.1** $\mathbf{X} = (X, \mathbb{X}) \in C^{p\text{-var}}([0,T];\mathcal{V}) \times C^{\frac{p}{2}\text{-var}}([0,T]^2; \mathrm{Sym}(\mathcal{V}\otimes\mathcal{V}))$ is said to be a *reduced rough path* if it satisfies the reduced Chen relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = \mathrm{Sym}(X_{s,u} \otimes X_{u,t}), \qquad \forall (s,u,t) \in [0,T]^3.$$

So long as one is willing to restrict attention to integrands with symmetric Gubinelli derivatives, one can define a theory of rough integration for reduced rough paths and Itô's formula in (4.2) still holds, as does a version of Theorem 3.1. The advantage of reduced rough paths is that their existence theory is trivial: given $S \in C^{p\text{-var}}([0,T];\mathcal{V})$ and $\gamma \in C^{\frac{p}{2}\text{-var}}([0,T];\mathrm{Sym}(\mathcal{V}\otimes\mathcal{V}))$, we may define a reduced rough-path with bracket $\gamma$ by

$$\mathbb{S}_{s,t} = \frac{1}{2}(S_{s,t} \otimes S_{s,t} + \gamma_{s,t}). \tag{4.4}$$

In particular, for any $S$ of finite $p$-variation for $p \in (2,3)$, we can find a reduced rough path $(S, \mathbb{S})$ satisfying (4.3), which then implies that (3.1) holds. Thus if the market prices for the hedging instruments all satisfy (4.1), we can replicate the payoff $f_T^0$ by charging $V_0^0$ and following the gamma-hedging strategy, where $V_t^0$ is the solution to (4.1) with the appropriate terminal condition.

Our treatment so far has not mentioned probability theory, but we now note that the pricing functions $V^i$ have a clear probabilistic interpretation: If $\tilde{S}$ follows a $d$-dimensional Itô diffusion given by (1.2), then by the Feynman–Kac theorem, if the $V^i$ are given by (1.1) and satisfy the conditions of Theorem 1.1, they satisfy (4.1). Combining these observations, we have proved Theorem 1.1 from the introduction.

### 4.1 Calibration to the Black–Scholes model

As an application of our results, we consider the options market as it existed before the stock market crash of October 1987. Before the crash, there was no pronounced

volatility smile, but it has subsequently become a persistent feature of the market; see Rubinstein [42, 43].

To apply our theory, we take $d = 2$ and $U = (0, \infty) \times (0, \infty)$. We use upper indices for the components of $S \in U \subseteq \mathbb{R}^2$. The component $S_t^1$ of our pricing signal $S$ is taken to give the current stock price. The component $S_t^2$ is assumed to give the common implied volatility of a set of $n$ call options on the stock; so we assume these prices exhibit no volatility smile. We suppose that all options have maturity $T$ and strikes $K^i$ $(1 \leq i \leq n)$. The payoffs of the options are given by $f^i(S_T) = \max\{S_T^1 - K^i, 0\}$.

Let $(\tilde{S}_t^{s_1,s_2,u})_{t\in[u,T]}$ be the $U$-valued process with components which satisfy the stochastic differential equations

$$d\tilde{S}_t^1 = \tilde{S}_t^1 \tilde{S}_t^2 \, dW_t^1,$$
$$d\tilde{S}_t^2 = 0$$

with initial condition $\tilde{S}_u = (s_1, s_2)$. The assumption that $S_t^2$ is the common implied volatility of the options means that we have assumed that at each time $t$, the prices of these options are given by $V^i(S_t^1, S_t^2, t)$, where

$$V^i(s_1, s_2, u) = \mathbb{E}\big[f^i\big((\tilde{S}^{s_1,s_2,u})_T^1\big)\big]. \tag{4.5}$$

This is the simplest form of calibration of an option pricing model, as we use only one calibration parameter $S^2$. Note that the assumption that $\tilde{S}^2$ is constant in the diffusion equation used for pricing does not prevent $S^2$ from changing when we come to apply our results.

The gamma and vega of European options in the Black–Scholes model with the same time to maturity $\tau$ do not need to be hedged separately due to the following result.

**Lemma 4.2** *Let* $\mathrm{BS}^f(s_1, s_2, \tau)$ *denote the Black–Scholes price of a European derivative with payoff* $f(x)$*, where* $x$ *is the stock price at maturity, the current stock price is* $s_1$*, the volatility is* $s_2$ *and the time to maturity is* $\tau$*. One then has*

$$\frac{\partial}{\partial s_2}\mathrm{BS}^f(s_1, s_2, \tau) = s_1^2 s_2 \tau \frac{\partial^2}{\partial s_1^2}\mathrm{BS}^f(s_1, s_2, \tau). \tag{4.6}$$

***Proof*** See the Appendix. □

Since the Black–Scholes formula is smooth for all $t < T$, one can view the options with maturity $T$ as European options with smooth payoffs at earlier times $t < T$ that depend on both parameters $(S_t^1, S_t^2)$. Theorem 1.1 then applies. By Lemma 4.2, the rank of the matrix

$$\left(\frac{\partial V^i}{\partial s_1}\bigg|_t, \frac{\partial V^i}{\partial s_2}\bigg|_t, \frac{\partial^2 V^i}{\partial s_1^2}\bigg|_t, \frac{\partial^2 V^i}{\partial s_1 \partial s_2}\bigg|_t, \frac{\partial^2 V^i}{\partial s_2^2}\bigg|_t\right)_{1\leq i\leq n}$$

is at most 4. Thus assuming that the signal $S$ has finite $p$-variation for $p < 3$ and the matrix has the maximal rank 4 for all times up to $t$, we can replicate any European derivative payoff $f(S_t^1)$ with $f$ smooth and $t < T$ by charging the Black–Scholes price $\mathrm{BS}^f(S_0^1, S_0^2, t)$ and then pursuing the full gamma-hedging strategy of matching all the first and second order partial derivatives of the price with respect to the underlying stock price and the implied volatility.

We call this the full gamma-hedging strategy by way of contrast with the classical gamma-hedging strategy where only the partial derivatives with respect to the underlying are matched. If we are willing to assume that $S^2$ is sufficiently regular, the classical gamma-hedging strategy suffices, as the next result shows.

**Theorem 4.3** *Suppose a European option on an underlying $S^1$ (with maturity $T$ and convex, nonlinear payoff $f^1(S_T^1)$) is traded at all times for the price determined by the Black–Scholes implied volatility $S_t^2$ as in (4.5). Suppose $S^1 \in C^{p\text{-var}}([0,T];\mathbb{R}^+)$, $S^2 \in C^{q\text{-var}}([0,T];\mathbb{R}^+)$ with $p < 3$, $q < 2$ and $\frac{1}{p} + \frac{1}{q} > 1$. Given a smooth function $f^0$, the payoff $f^0(S_t^1)$ can be replicated at the Black–Scholes price for any $t \in [0,T)$ in the sense that the profit or loss of the discrete-time classical gamma-hedging strategy in the underlying and the option with payoff $f^1$ tends to zero as the mesh size of the partition $\pi$ tends to zero. Here the classical gamma-hedging strategy is to be understood as matching only the partial derivatives with respect to the underlying, i.e., only requiring (1.3) and (1.4) to hold for $\alpha = \beta = 1$. If $(f^1)''(s_1) > 0$ for all $s_1 > 0$, then it is possible to replicate the payoff $f^0$ at the maturity $T$.*

***Proof*** We label the option with payoff $f^1$ as instrument 1 and the stock as instrument 2 in order to apply the notation from the rest of the paper.

The purpose of our convexity assumption is to ensure that the Black–Scholes gamma of the option $f^1$ is positive at times $t < T$; see Lemma A.1 in the Appendix for a proof. Because the gamma of a stock is always 0 and its delta is always 1, the matrix

$$\begin{pmatrix} \frac{\partial V^1}{\partial s_1}\Big|_t & \frac{\partial^2 V^1}{\partial s_1^2}\Big|_t \\ \frac{\partial V^2}{\partial s_1}\Big|_t & \frac{\partial^2 V^2}{\partial s_1^2}\Big|_t \end{pmatrix}$$

has rank 2 at all times $t \in [0,T)$. Hence we can find bounded quantities $q_t^1$, $q_t^2$ such that

$$\sum_{i=0}^{2} q_t^i \frac{\partial V^i}{\partial s_1}\Big|_t = 0, \qquad \sum_{i=0}^{2} q_t^i \frac{\partial^2 V^i}{\partial s_1^2}\Big|_t = 0,$$

with $q_t^0 = -1$ for all times $t \in [0,T)$. If we make the additional assumption that $(f^1)''(s^1) > 0$ for all $s^1 > 0$, we can find such $q_t^i$ for all $t \in [0,T]$.

Our first aim is to write an integral formula for the price using a rough-path integral in $S^1$ and a Young integral in $S^2$. We achieve this in (4.8) below. However, to understand this equation, we need to introduce a slightly more flexible definition of a controlled rough path to ensure that the integrand is controlled.

Let $(X, \mathbb{X}) \in \mathcal{C}^{p\text{-var}}([0,T];\mathcal{V})$ be given. Let $r$ be chosen such that $\frac{1}{r}+\frac{2}{p} > 1$. Suppose that $(Y, Y') \in C^{p\text{-var}}([0,T];\hat{\mathcal{V}}) \times C^{r\text{-var}}([0,T];\mathcal{L}(\mathcal{V};\hat{\mathcal{V}}))$ and that the remainder $R^Y : [0,T]^2 \to \hat{\mathcal{V}}$ by $R^Y_{s,t} := Y_{s,t} - Y'_s X_{s,t}$ has finite $(\frac{1}{p}+\frac{1}{r})^{-1}$-variation. Then we say that $(Y, Y')$ is $X$-controlled with $(p,r)$-variation regularity. The case $r = p$ corresponds to the notion of controlled rough path we have used previously. We can define a rough integral for such paths and Theorem 2.3 still holds with an appropriate choice for the control function; see Armstrong et al. [4].

For our application, we want to view the price process as a controlled rough path with Gubinelli derivative given by the delta. We know from the theory when $r = p$ that the remainder $R^{V^i} : [0,T]^2 \to \mathbb{R}$ given by

$$R^{V^i}_{s,t} := V^i_{s,t} - \left.\frac{\partial V^i}{\partial s_1}\right|_s S^1_{s,t} - \left.\frac{\partial V^i}{\partial s_2}\right|_s S^2_{s,t}$$

has finite $\frac{p}{2}$-variation regularity. The third term has finite $q$-variation regularity. Hence $(V^i, \frac{\partial V^i}{\partial s_1})$ is an $S^1$-controlled rough path of $(p,r)$-variation regularity so long as

(i) $\frac{1}{r}+\frac{2}{p} > 1$ (from the definition of $(p,r)$-variation regularity);
(ii) $\frac{1}{p} \geq \frac{1}{r}$ (to ensure $\frac{\partial V^i}{\partial s_1} \in C^{r\text{-var}}([0,T];\mathbb{R})$);
(iii) $\frac{1}{q} \geq \frac{1}{p}+\frac{1}{r}$ and $\frac{2}{p} \geq \frac{1}{p}+\frac{1}{r}$ (to ensure $R^{V^i}$ is sufficiently regular).

These conditions are equivalent to $\frac{1}{p} \geq \frac{1}{r} > 1 - \frac{2}{p}$ and $\frac{1}{q} - \frac{1}{p} \geq \frac{1}{r}$. So we can find such an $r$ if $\frac{1}{q} - \frac{1}{p} > 1 - \frac{2}{p}$, equivalently if $\frac{1}{p}+\frac{1}{q} > 1$. Hence we may choose $r$ to ensure that $(V^i, \frac{\partial V^i}{\partial s_1})$ is an $S^1$-controlled rough path of $(p,r)$-variation regularity.

Choose the reduced-rough-path lift $\mathbb{S}$ so that $\mathbf{S} := (S, \mathbb{S})$ satisfies

$$[\mathbf{S}]_t = \begin{pmatrix} [\mathbf{S}^1]_t & 0 \\ 0 & 0 \end{pmatrix},$$

where $\mathbf{S}^1$ is the reduced-rough-path lift of $S^1$ given by requiring

$$[\mathbf{S}^1]_t = \int_0^t (S^1_u S^2_u)^2 \, du.$$

By the rough-path Itô formula, we have

$$\begin{aligned} V^i(S^1_t, S^2_t, t) = V^i(S^1_0, S^2_0, 0) &+ \int_0^t \left.\frac{\partial V^i}{\partial t}\right|_u du \\ &+ \left( \left( \begin{pmatrix} \frac{\partial V^i}{\partial s_1} \\ \frac{\partial V^i}{\partial s_2} \end{pmatrix}, \begin{pmatrix} \frac{\partial^2 V^i}{\partial s_1^2} & \frac{\partial^2 V^i}{\partial s_1 \partial s_2} \\ \frac{\partial^2 V^i}{\partial s_1 \partial s_2} & \frac{\partial^2 V^i}{\partial s_2^2} \end{pmatrix} \right) \cdot (S, \mathbb{S}) \right)_{0,t} \\ &+ \int_0^t \frac{1}{2} \left.\frac{\partial^2 V^i}{\partial s_1^2}\right|_u d[\mathbf{S}^1]_u \end{aligned} \tag{4.7}$$

with the final integral being a Young integral. Using (4.4), we may write the lift $\mathbb{S}$ as

$$\mathbb{S}_{s,t} = \frac{1}{2}(S^1_{s,t} \odot S^1_{s,t} + 2S^1_{s,t} \odot S^2_{s,t} + S^2_{s,t} \odot S^2_{s,t}) + [\mathbf{S}]_{s,t},$$

where $\odot$ denotes the symmetric tensor product. Our hypothesis that $\frac{1}{p} + \frac{1}{q} > 1$ together with Lemma 2.2 allows us to write

$$|S^1_{s,t} \odot S^2_{s,t}| \leq \|S^1\|_{p\text{-var};[s,t]} \|S^2\|_{q\text{-var};[s,t]} \leq w(s,t)^{1+\epsilon}$$

for some control function $w$ and $\epsilon > 0$. Since we also assume $q < 2$, we can bound the square of $S^2_{s,t}$ in the same way. Hence if we use the definition of the rough integral to write (4.7) as the limit of compensated Riemann sums, all the second-order derivative terms involving differentiation by $s_2$ vanish by (2.1) as the mesh size of $\pi$ tends to 0. It therefore follows from (4.7) that

$$\begin{aligned} V^i(S^1_t, S^2_t, t) = V^i(S^1_0, S^2_0, 0) &+ \int_0^t \left.\frac{\partial V^i}{\partial t}\right|_u du \\ &+ \int_0^t \left.\frac{\partial V^i}{\partial s_1}\right|_u d\mathbf{S}^1_u + \int_0^t \left.\frac{\partial V^i}{\partial s_2}\right|_u dS^2_u + \int_0^t \frac{1}{2}\left.\frac{\partial^2 V^i}{\partial s_1^2}\right|_u d[\mathbf{S}^1]_u, \end{aligned} \tag{4.8}$$

where the first integral is a Riemann integral, the second is a rough integral and the third and fourth are Young integrals. This is because if one writes out the integrals in (4.7) and (4.8) in terms of their standard expressions as limits of compensated Riemann sums (see (2.2) and [22, Exercise 6.9]), they only differ by the vanishing terms. Since the $V^i$ satisfy the Black–Scholes PDE, we obtain the formula

$$V^i_t = V^i_0 + \int_0^t \left.\frac{\partial V^i}{\partial s_2}\right|_u dS^2_u + \left(\left(\frac{\partial V^i}{\partial s_1}, \frac{\partial^2 V^i}{\partial s_1^2}\right) \cdot (S^1, \mathbb{S}^1)\right)_{0,t}, \tag{4.9}$$

with the first integral being a Young integral. By (4.6), we know that

$$\sum_{i=0}^{2} q^i_t \left.\frac{\partial V^i}{\partial s_2}\right|_t = 0$$

at all times. Hence we can apply the same argument used to prove Theorem 3.1 to prove our result. Since we have (4.9) in place of (3.1), two modifications are needed. First, when estimating the rough-path terms, we need to use the control function arising from the version of Theorem 2.3 for controlled rough paths of $(p, r)$-variation regularity. Second, we need to bound a Young integral where we previously bounded

a Riemann integral. The additional estimate needed for the Young integral terms is

$$\begin{aligned}\left|\sum_{i=0}^{n} q_s^i \int_s^t \frac{\partial V^i}{\partial s_2}\Big|_u \, dS_u^2\right| &\leq \left|\sum_{i=0}^{n} q_s^i \int_s^t \frac{\partial V^i}{\partial s_2}\Big|_u \, dS_u^2 - \sum_{i=0}^{n} q_s^i \frac{\partial V^i}{\partial s_2}\Big|_s S_{s,t}^2\right| \\ &\leq \sum_{i=0}^{n} |q_s^i| \left| \int_s^t \frac{\partial V^i}{\partial s_2}\Big|_u \, dS_u^2 - \frac{\partial V^i}{\partial s_2}\Big|_s S_{s,t}^2\right| \\ &\leq C \sum_{i=0}^{n} |q_s^i| \left\| \frac{\partial V}{\partial s_2} \right\|_{p\text{-var};[s,t]} \|S^2\|_{q\text{-var};[s,t]},\end{aligned}$$

where the last line uses the Young–Loève estimate; see [22, Theorem 6.8]. Applying Lemma 2.2 to the $p$- and $q$-variation controls together with the condition $\frac{1}{p} + \frac{1}{q} > 1$ allows us to write

$$\left|\sum_{i=0}^{n} q_s^i \int_s^t \frac{\partial V^i}{\partial s_2}\Big|_u \, dS_u^2\right| \leq C w(s,t)^{1+\epsilon}$$

for some control $w(s,t)$ and $\epsilon > 0$. Hence we may apply the estimate (2.1) to find

$$\sum_{[s,t]\in\pi} \left|\sum_{i=0}^{n} q_s^i \int_s^t \frac{\partial V^i}{\partial s_2}\Big|_u \, dS_u^2\right| \longrightarrow 0 \qquad \text{as } |\pi| \to 0.$$

This allows us to prove the appropriate analogue of Theorem 3.1, and hence we obtain the desired result. □

We remark that a general pattern has emerged: the same estimates used to show the convergence of integrals written as Riemann sums allow us to prove replication results.

Our results above show that in a market with no volatility smile, we can replicate an infinite-dimensional family of payoff functions using only a finite number of derivative instruments. It is impossible to do this by using a static replication strategy.

Our strategy does not require the trader to use any information about the paths $S^1$ and $S^2$ beyond what is known at time $t$ and the assumed regularity. The trader's strategy computes the Greeks using the classical constant-volatility Black–Scholes pricing formulae with the instantaneous implied volatility $S_t^2$ as input.

In this example, replication is possible even though the probability model is not determined at a time $u \in [0,T)$ by the requirement that $S_u^2$ equals the implied volatility of the hedging options. At each time $u \in [0,T)$, there is an infinite family of non-equivalent arbitrage-free $\mathbb{Q}$-measure models compatible with this requirement and our regularity assumptions: one can use any model of the form

$$d\tilde{S}_t^1 = \tilde{S}_t^1 v_t \, dW_t, \qquad \tilde{S}_u^1 = S_u^1,$$

$$\tilde{S}_t^2 = \sqrt{\frac{1}{T-t}\int_t^T v_s^2 \, ds}$$

with $v \in C^\infty([u, T]; (0, \infty))$ chosen to ensure that

$$\tilde{S}_u^2 = S_u^2.$$

This is because as is well known, the implied volatility of a European option in such a time-dependent volatility model is equal to the root-mean-square volatility up to maturity.

We find it interesting to note that this hedging methodology also works even if the realised signal cannot arise from any classical arbitrage-free pricing model. It is true that if one knows the signal comes from a specific $\mathbb{P}$-measure model which contains arbitrage, one can exploit this to replicate anything. However, in reality, the "true" dynamics are unknown and this may prevent such an arbitrage being exploited. Even if the dynamics were known, it might be infeasible to compute an arbitrage strategy. Moreover, the notion that there really are "true" dynamics underpinning the market is highly debatable. The gamma-hedging strategy remains effective despite these issues.

The discussion above explains in part why we believe it is interesting to give a rough-path theory proof of the effectiveness of our strategy rather than a probabilistic proof using semimartingales. Another reason is that classical arbitrage-free pricing models do not typically consider market impacts. The gamma-hedging strategy remains effective even in the extreme case that an adversary can select the path taken by the signal in full knowledge of the strategy that we pursue. It also remains robust to more realistic forms of price impact.

Let us conclude this section with a few remarks on the economic interpretation of these results and their comparison with the Black–Scholes model.

The classical replication result of Black–Scholes shows the effectiveness of the sell-side strategy of delta hedging. In the Black–Scholes model, a trader can replicate options that buy-side investors wish to purchase by charging the Black–Scholes price. If the trader charges a little more than that, they can make a risk-free profit, subject to the assumption that the stock price process adheres to the Black–Scholes model.

Our result is similar except that rather than assuming properties of the dynamics of the stock price, the trader makes assumptions on certain non-probabilistic properties of exchange-traded option prices. The validity of either assumption is an empirical question. In the case of the pre-October 1987 market, there is good historical evidence that the volatility smile was approximately flat. Our approach seems to apply well to this historical market. On the other hand, as early as 1963, Mandelbrot reported evidence that return distributions deviated significantly from the predictions of the Black–Scholes model; see Mandelbrot [36].

In both replication approaches, for the result to be financially useful, it is important that the replication error remains small under small perturbations of the modelling assumptions. In the case of the classical Black–Scholes approach, this is established by the fundamental theorem of derivative trading. In our approach, one can appeal to the good continuity properties of rough differential equations to obtain a similar result; see Ionescu [31, Theorem 14.3.2] for details.

Girsanov's theorem shows that the physical measure of Black–Scholes theory is equivalent to a martingale measure and hence the model admits no arbitrage. In our model, the formal definition of arbitrage does not apply as our model does not assume the existence of a physical probability measure and this features in the definition of

arbitrage. Nevertheless, clearly one should require our pricing model to be consistent in some sense; for example, there should be no violation of put-call parity. Our pricing model is consistent in this sense as there always exists at least one classical arbitrage-free physical measure $\mathbb{P}$, namely $\mathbb{P} = \mathbb{Q}$, which gives the same prices.

There are many other issues one should consider when implementing a sell-side strategy. These include the practicalities of calibrating pricing models versus statistical models, the impact of transaction costs and market jumps. Our approach should be viewed as a simple theoretical model which is compatible with the practices of calibrating to market data and gamma hedging.

We have considered the historical options market as an example because of its simplicity, and because of the straightforward empirical evidence on the shape of the volatility smile at this time. In contemporary markets, one might instead assume that a more sophisticated $\mathbb{Q}$-measure model with a curved volatility smile can be used to price the chosen hedging instruments to obtain similar results.

## 5 Path-dependent instruments

### 5.1 Controllable payoffs

To apply Theorem 3.1 to path-dependent derivatives, we need to identify circumstances under which the price of a derivative in the diffusion model (1.2) obeys the rough-path integral formula (3.1). We assume **from this point forward** that the matrix $\sigma(s,t)$ is invertible for all $s \in U$ and $t \in [0,T]$. We also assume that the drift $\mu(s,t)$ is zero.

If $G \in L^2(\Omega)$ represents the payoff of a derivative and $V_t^G = \mathbb{E}_t[G]$ is its price at time $t$, then by the martingale representation theorem and our assumption on the invertibility of $\sigma$, we may define $\Delta^G$ to be the $\mathcal{L}(\mathbb{R}^d;\mathbb{R})$-valued predictable process such that

$$V_t^G = V_0^G + \int_0^t \Delta_u^G \, d\tilde{S}_u, \tag{5.1}$$

where the integral is an Itô integral and $\tilde{S}$ is the process defined by (1.2). We should like to consider when this Itô integral may be written as a rough-path integral. Let us summarise the relevant rough-path theory (see Friz and Hairer [21, Chap. 5]).

Recall that the scalar Itô integral is defined using a limit of sums of the scalar product of the integrand with differences in the integrator. If one modifies this definition to use a tensor product, one can define a tensor Itô integral in exactly the same way. We use this integral in our next definition.

**Definition 5.1** Given a local martingale $M$, we define its *Itô enhancement* by the Itô integral

$$\mathbb{M}_{s,t}^{\text{Itô}} = \int_s^t M_{s,u} \otimes dM_u.$$

We may now describe the sense in which Itô and rough-path integrals coincide.

**Proposition 5.2** *Let $M$ be a local martingale with $M_t = M_0 + \int_0^t \phi_u \, dW_u$ for $t \in [0, T]$. Assume that $(M(\omega), \phi(\omega)) \in \mathcal{D}^{p\text{-var}}_{W(\omega)}$ for almost all $\omega \in \Omega$. Suppose we have $(Y(\omega), Y'(\omega)) \in \mathcal{D}^{p\text{-var}}_{M(\omega)}$ for almost all $\omega \in \Omega$. If $Y$, $Y'$ are adapted, then almost surely,*

$$\int_0^t Y_u \, dM_u = \left((Y, Y') \cdot (M, \mathbb{M}^{It\hat{o}})\right)_{0,t}.$$

*where the left integral is an Itô integral.*

The result is well known. We give a proof in Appendix for completeness. The result motivates the following definition of *controllable payoffs*.

**Definition 5.3** Let $\tilde{S}$ be a path of finite $p$-variation. Then $G \in L^0(\Omega; \mathbb{R})$ is called an *$\tilde{S}$-controllable payoff of $p$-variation regularity* if $\Delta^G$ almost surely admits a Gubinelli derivative $\Gamma^G$ with respect to $\tilde{S}$ equipped with its Itô enhancement. That is, for almost all $\omega \in \Omega$, we have

$$\left(\Delta^G(\omega), \Gamma^G(\omega)\right) \in \mathcal{D}^{p\text{-var}}_{S(\omega)}$$

where $\Delta^G(\omega)(t) := \Delta^G_t(\omega)$ and $\Gamma^G(\omega)(t) := \Gamma^G_t(\omega)$.

Suppose in addition that $G$ is continuous in the sup-norm when we choose $\Omega$ as the space of continuous paths of Brownian motion starting at 0. Suppose also that there is a continuous map

$$\begin{aligned}\zeta : &C^{p\text{-var}}([0, T]; \mathbb{R}^d) \\ &\to C^{p\text{-var}}([0, T]; \mathbb{R}^d) \oplus C^{p\text{-var}}([0, T]; \mathbb{R}^{d^2}) \oplus C^{\frac{p}{2}\text{-var}}([0, T]^2; \mathbb{R}^d)\end{aligned}$$

satisfying $\zeta(\omega) = (\Delta^G(\omega), \Gamma^G(\omega), R^G(\omega))$ almost surely, where we define the remainder by $R^G(\omega)(s, t) = \Delta^G_t - \Delta^G_s - \Gamma^G_{s,t}\tilde{S}_s$. In these circumstances, we say that $G$ is *continuously controlled*.

We now note the following lemma.

**Lemma 5.4** *Under our assumption that $\sigma$ is invertible, $G$ is $\tilde{S}$-controllable if and only if it is $W$-controllable.*

A proof is given in the Appendix as Lemma A.4. Our proof contains an explicit formula for the relationship between the Greeks in this situation. As a result of this lemma, we may focus our attention on $W$-controllable payoffs henceforth.

When all the hedging instruments and the instrument being replicated are $W$-controllable, we know that (3.1) holds almost surely. Hence by Theorem 3.1, the error of discrete-time gamma hedging on a grid $\pi$ almost surely tends to 0 as $|\pi| \to 0$ so long as all the payoffs are $W$-controllable. This in itself is a rather weak result; there are already known results showing that the delta-hedging strategy converges almost surely for appropriately chosen sequences $(\pi_n)_{n \in \mathbb{N}}$ of

partitions with mesh size tending to zero, albeit not for all possible choices (see Bick and Willinger [8]). However, we now consider how to use the continuity of the rough-path integral in its integrand and integrator in the appropriate rough-path topologies to obtain a sure result for continuously controlled payoffs.

A smooth path $X \in C^\infty([0,T];\mathbb{R}^d)$ has a canonical lift $\mathbf{X} = (X, \mathbb{X}^G)$ given by

$$\mathbb{X}^G_{s,t} = \int_s^t X_{s,r} \otimes dX_r,$$

called the *geometric lift* of $X$. However, this is not the only choice of lift; we may define its *Brownian lift* $\mathbf{X}^B = (X, \mathbb{X}^B)$ by

$$\mathbb{X}^B_{s,t} = \mathbb{X}^G_{s,t} - \frac{1}{2}(t-s)I,$$

where $I$ is the identity matrix. We call this the Brownian lift because its rough bracket $[\mathbf{X}]$ is equal to that of Itô-enhanced Brownian motion. We may define the Brownian lift of piecewise smooth paths in the same way.

**Definition 5.5** For $p \in (2,3)$, we define $\mathcal{B}^p([0,T];\mathbb{R}^d)$, which we call *the space of pseudo-Brownian paths*, to be the closure of

$$\{(X, \mathbb{X}^B) : X \in C^\infty([0,T];\mathbb{R}^d),\, X_0 = 0\}$$

in the rough-path $p$-variation norm.

By [21, Exercise 2.9], piecewise linear paths equipped with the Brownian lift lie in $\mathcal{B}_p$. By [21, Proposition 3.6], dyadic piecewise linear approximations to Brownian motion equipped with the geometric lift converge to Brownian motion with the geometric lift. Hence Brownian motion with its Itô lift almost surely lies in $\mathcal{B}_p$.

The continuity properties of the rough integral allow us to prove sure results because, as the next result asserts, the set of paths of the form $\mathbf{W}^{\mathrm{Itô}} = (W, \mathbb{W}^{\mathrm{Itô}})$ is dense in $\mathcal{B}^p([0,T];\mathbb{R}^d)$.

**Proposition 5.6** *Suppose* $\tilde{\mathbf{W}} \in \mathcal{B}^p([0,T];\mathbb{R}^d)$. *Then for any* $\epsilon > 0$,

$$\mathbb{Q}[|||\mathbf{W}^{Itô} - \tilde{\mathbf{W}}|||_{p\text{-var};[0,T]} < \epsilon] > 0,$$

*where* $\mathbb{Q}$ *is the measure introduced before Theorem* 1.1

***Proof*** See the Appendix. □

While conditional expectations are usually only defined almost surely, we now explain how to define expectations conditioned on $\mathcal{F}_t$ surely. Given a continuous path $\tilde{W}$ starting at 0 and defined on an interval $[0, t_0]$, we may define a probability space consisting of paths which follow $\tilde{W}$ up to time $t_0$ and Brownian motion thereafter. We define $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbb{Q}})$ by taking $\tilde{\Omega}$ to be the set of continuous paths on $[t_0, T]$ starting at $\tilde{W}_{t_0}$ and choosing $\tilde{\mathcal{F}}$ and $\tilde{\mathbb{Q}}$ such that if $\tilde{W} \in \tilde{\Omega}$ then $(\tilde{W}_{t+t_0} - \tilde{W}_{t_0})$ is a Brownian

motion on $[0, T - t_0]$ generating $\tilde{\mathcal{F}}$. Given a measurable function $G : \Omega \to \mathbb{R}$, we define

$$\mathbb{E}[G|(\tilde{W}_t)_{t\in[0,t_0]}] := \mathbb{E}_{\tilde{\mathbb{Q}}}[G(\tilde{W}_\cdot)].$$

This is defined surely and is almost surely equal to the conditional expectation in the sense that

$$\mathbb{E}[G|\mathcal{F}_{t_0}](W_\cdot) = \mathbb{E}[G|(W_t)_{t\in[0,t_0]}] \qquad \text{a.s.}$$

Using this definition for conditional expectations, we can prove a rough-path Clark–Ocone formula.

**Theorem 5.7** *Suppose that $G$ is continuously $W$-controllable. If $t_0 \in [0, T]$ and $\tilde{\mathbf{W}} = (\tilde{W}, \tilde{\mathbb{W}}) \in \mathcal{B}^p([0, T]; \mathbb{R}^d)$, then*

$$\mathbb{E}[G|(\tilde{W}_t)_{t\in[0,t_0]}] = \mathbb{E}[G] + \Big(\big(\Delta^G(\tilde{W}), \Gamma^G(\tilde{W})\big) \cdot (\tilde{W}, \tilde{\mathbb{W}})\Big)_{0,t_0}.$$

***Proof*** By the definition of $\Delta^G$, we know that

$$\mathbb{E}[G|\mathcal{F}_{t_0}] = \mathbb{E}[G] + \int_0^{t_0} \Delta_u^G \, dW_u$$

for all $t_0$, and so we also have that

$$G = \mathbb{E}[G|\mathcal{F}_{t_0}] + \int_{t_0}^{T} \Delta_u^G \, dW_u.$$

Hence

$$G(W_\cdot) = \mathbb{E}[G|(W_t)_{t\in[0,t_0]}] + \Big(\big(\Delta^G(W), \Gamma^G(W)\big) \cdot (W, \mathbb{W})\Big)_{t_0,T} \tag{5.2}$$

almost surely by the equivalence of Itô and rough integrals. The continuity of $R^G$ ensures that $\Gamma^G$ is surely a Gubinelli derivative for $\Delta^G$, and hence the rough integral surely exists if $\tilde{\mathbf{W}} \in \mathcal{B}^p([0, T]; \mathbb{R}^d)$. Then by the continuity of all the terms and Proposition 5.6, (5.2) holds surely for $\tilde{\mathbf{W}} \in \mathcal{B}^p([0, T]; \mathbb{R}^d)$. By equating (5.2) with the same equation when $t_0 = 0$ and simplifying, the result follows. □

**Corollary 5.8** *Suppose $\tilde{W}$ is a path starting at 0 with finite $p$-variation, $2 < p < 3$. If $G^i$ for $0 \le i \le n$ are continuously $W$-controllable payoffs and if derivatives with payoffs $G^i$ $(1 \le i \le n)$ can be purchased at each time $t_0$ for the price*

$$\mathbb{E}[G^i|(\tilde{W}_t)_{t\in[0,t_0]}],$$

*then the profit or loss of the discrete-time gamma-hedging strategy to replicate $G^0$ using these derivatives tends to 0 as the mesh size of the partition tends to 0.*

***Proof*** Let $C_0^{0,p\text{-var}}$ denote the closure in the $p$-variation norm of the set of smooth paths starting at 0 and $C_0^{p\text{-var}}$ the set of all paths of finite $p$-variation starting at 0. The result for $W \in C_0^{0,p\text{-var}}$ is an immediate consequence of Theorem 3.1 and Theorem 5.7. If $2 < q < p$, then $C_0^{0,q\text{-var}} \subseteq C_0^{p\text{-var}}$ by [22, Corollary 5.35]. This gives the result. □

In order to identify the Gubinelli derivatives of Itô integrals, we make frequent use of the following result, which shows that the Itô integral is regularising.

**Proposition 5.9** *Let $W$ be a $d$-dimensional Brownian motion and $\phi$ an $\mathbb{R}^{m\times d}$-valued continuous adapted process satisfying $\mathbb{E}[\int_0^T (\phi_s\phi_s^\top)_{ij}\,ds] < \infty$ for all components $i, j = 1, \dots, m$. Set $X_t = \int_0^t \phi_u\,dW_u$ for all $t \in [0, T]$. Suppose almost all sample paths of $\phi$ are of finite $p$-variation. Then with probability one, we have*

(i) $X \in C^{p\text{-var}}([0, T]; \mathbb{R}^m)$;
(ii) $(X, \phi) \in \mathcal{D}_W^{p\text{-var}}$, *i.e., the remainder $R^X$ defined as*

$$R^X_{s,t} = X_{s,t} - \phi_s W_{s,t} = \int_s^t \phi_{s,u}\,dW_u$$

*is of finite $\frac{p}{2}$-variation*;
(iii) *for any $f \in C_b^3(\mathbb{R}^d; \mathbb{R}^{m\times d})$, $(f(X), \nabla f(X)\phi)$ belongs to $\mathcal{D}_W^{p\text{-var}}$.*

We do not claim this result is new, but as we could not find a proof of the second part in the literature, we give a proof in the Appendix.

## 5.2 Examples

### 5.2.1 Simple payoffs

Let $C^\infty_{\text{poly}}(\mathbb{R}^n; \mathbb{R})$ denote the space of smooth functions with all derivatives of polynomial growth.

**Definition 5.10** For each $h \in C^\infty([0, T]; \mathbb{R}^d)$, define $\mathcal{W}(h) \in \mathbb{R}$ by

$$\mathcal{W}(h) := h_T W_T - h_0 W_0 - \int_0^T W_t\,dh_t = \int h_t\,dW_t \qquad \text{a.s.}$$

The set of all payoffs of the form

$$G = f\big(\mathcal{W}(h_1), \dots, \mathcal{W}(h_n)\big),$$

where $h_1, \dots, h_n \in C^\infty([0, T]; \mathbb{R}^d)$ and $f \in C^\infty_{\text{poly}}(\mathbb{R}^n; \mathbb{R})$, is the space of *simple payoffs with maturity $T$*. We denote this space by $\mathcal{S}^\infty$.

We have defined $\mathcal{W}(h)$ above as a Young integral with continuous integrand, but note that it is almost surely equal to an Itô integral. The Young-integral formulation makes it clear that $G$ is defined surely on $\Omega$ and is continuous in the sup-norm.

**Theorem 5.11** *Each $G \in \mathcal{S}^\infty$ is continuously $W$-controllable with*

$$\Delta_u^G = \mathbb{E}_u[D_u G] \quad \textit{and} \quad \Gamma_u^G = \mathbb{E}_u[D_u \Delta_u^G], \qquad u \in [0, T],$$

*when the delta and gamma are computed in the Bachelier model $\tilde{S} = W$ and $D_u$ denotes the Malliavin derivative evaluated at time $u$.*

***Proof*** For brevity, we set $X = (\mathcal{W}(h_1), \ldots, \mathcal{W}(h_n))$. By the definition of the Malliavin derivative,

$$D_u G = \sum_{i=1}^n \frac{\partial f(X)}{\partial x_i} h_i(u).$$

By the Clark–Ocone formula,

$$\Delta_u^G = \sum_{i=1}^n h_i(u) \mathbb{E}_u\left[\frac{\partial f(X)}{\partial x_i}\right].$$

Hence

$$\mathbb{E}_u[D_u \Delta_u^G] = \sum_{1 \le i,j \le n} \big(h_i(u) \otimes h_j(u)\big) \mathbb{E}_u\left[\frac{\partial^2 f(X)}{\partial x_i \partial x_j}\right]. \tag{5.3}$$

Applying Clark–Ocone to each $\frac{\partial^2 f(X)}{\partial x_i \partial x_j}$, $1 \le i, j \le n$, we have

$$\mathbb{E}_u\left[\frac{\partial^2 f(X)}{\partial x_i \partial x_j}\right] = \mathbb{E}\left[\frac{\partial^2 f(X)}{\partial x_i \partial x_j}\right] + \int_0^u \left(\sum_{k=1}^n h_k(s) \mathbb{E}_s\left[\frac{\partial^3 f(X)}{\partial x_i \partial x_j \partial x_k}\right]\right) dW_s$$

for all $u \in [0, T]$. By Proposition 5.9,

$$\left\{\mathbb{E}_u\left[\frac{\partial^2 f(X)}{\partial x_i \partial x_j}\right] : u \in [0, T]\right\} \in C^{p\text{-var}}([0, T]; \mathbb{R})$$

almost surely. Also, $h_i \otimes h_j \in C^\infty([0, T]; \mathbb{R}^{d \times d})$ for all $i, j$. Again by Proposition 5.9 and (5.3), we deduce that $\{\mathbb{E}_u[D_u \Delta_u^G] : u \in [0, T]\}$ belongs to $C^{p\text{-var}}([0, T]; \mathbb{R}^d)$

almost surely. For the remainder term, let $(u, v) \in \Delta_T$. We have that

$$\begin{aligned}
R^G_{u,v} &:= \Delta^G_v - \Delta^G_u - W_{u,v}\mathbb{E}_u[D_u\Delta^G_u] \\
&= \sum_{i=1}^n \left( h_i(v)\mathbb{E}_v\left[\frac{\partial f(X)}{\partial x_i}\right] - h_i(u)\mathbb{E}_u\left[\frac{\partial f(X)}{\partial x_i}\right]\right) \\
&\quad - \sum_{i=1}^n\sum_{j=1}^n \big(h_i(u)\otimes h_j(u)\big)W_{u,v}\mathbb{E}_u\left[\frac{\partial^2 f(X)}{\partial x_i \partial x_j}\right] \\
&= \sum_{i=1}^n h_i(u)\left(\mathbb{E}_v\left[\frac{\partial f(X)}{\partial x_i}\right] - \mathbb{E}_u\left[\frac{\partial f(X)}{\partial x_i}\right]\right) \\
&\quad - \sum_{i=1}^n\sum_{j=1}^n \big(h_i(u)\otimes h_j(u)\big)W_{u,v}\mathbb{E}_u\left[\frac{\partial^2 f(X)}{\partial x_i \partial x_j}\right] \\
&\quad + \sum_{i=1}^n \big(h_i(v) - h_i(u)\big)\mathbb{E}_v\left[\frac{\partial f(X)}{\partial x_i}\right] \\
&= \sum_{i=1}^n h_i(u)\int_u^v \sum_{j=1}^n \left( h_j(s)\mathbb{E}_s\left[\frac{\partial^2 f(X)}{\partial x_i \partial x_j}\right] - h_j(u)\mathbb{E}_u\left[\frac{\partial^2 f(X)}{\partial x_i \partial x_j}\right]\right) dW_s \\
&\quad + \sum_{i=1}^n \big(h_i(v) - h_i(u)\big)\mathbb{E}_v\left[\frac{\partial f(X)}{\partial x_i}\right],
\end{aligned}$$

where we have used the Clark–Ocone formula for $\frac{\partial f(X)}{\partial x_i}$, $1 \le i \le n$, in the last line. By Proposition 5.9, we deduce that $R^G \in C^{\frac{p}{2}\text{-var}}([0,T]^2;\mathbb{R})$ and thus that $\Gamma^G_u = \mathbb{E}_u[D_u\Delta^G_u]$ as claimed. □

Note that the simple payoffs are dense in $L^2(\Omega)$.

Recall that the Malliavin derivative (see Nualart [37, Chap. 1]) is defined by first defining its action on simple payoffs. This gives an unbounded, but closable operator. One uses the $L^{\tilde{p}}$-closure of its graph to obtain for any $\tilde{p} \ge 1$ the space $\mathbb{D}^{1,\tilde{p}}$ of Malliavin differentiable functions. Thus by relating the delta and gamma of simple payoffs to the Malliavin derivative, we have obtained a fundamental connection between Malliavin derivatives and the delta and gamma for any payoffs that can be well approximated in both Malliavin and rough-path topologies. This idea is pursued further in [31, Proposition 13.2.18].

Note that the value of the Gubinelli derivative $\Gamma_t$ is a symmetric tensor for simple payoffs. As a result, it is possible to replicate simple payoffs by gamma-hedging strategies in European options.

### 5.2.2 Asian options in the Bachelier model

Consider payoffs of the form $G = f(A_T)$, where $A_T = \frac{1}{T}\int_0^T W_u\,du$ is the arithmetic average of the stock price over the lifetime of the option and $f$ is smooth. By Itô's

lemma applied to $(tW_t)_{t\in[0,T]}$, we see that

$$\int_0^T W_u\,du = -\int_0^T u\,dW_u + TW_T = \int_0^T (T-u)\,dW_u$$

so that $A_T = \int_0^T (1-\frac{u}{T})\,dW_u$. Using Theorem 5.11 with $h(u) = 1-\frac{u}{T}$, we conclude that $G = f(A_T)$ is smoothly $W$-controllable. Moreover, the Greeks are given by the expressions $\Delta_u^G = (1-\frac{u}{T})\mathbb{E}_u[f'(A_T)]$ and $\Gamma_u^G = (1-\frac{u}{T})^2\mathbb{E}_u[f''(A_T)]$.

### 5.2.3 Signature payoffs

We call a row vector $\alpha = (j_1, \dots, j_\ell)$, where $j_i \in \{0, 1, \dots, d\}$ for $i \in \{1, 2, \dots, \ell\}$, a *multi-index* of length $\ell := \ell(\alpha) \in \mathbb{N}$. Denote by $n(\alpha)$ the number of components of $\alpha$ which are equal to zero and by $\mathcal{M}$ the set of all multi-indices. That is,

$$\mathcal{M} := \left\{(j_1, \dots, j_\ell) : j_i \in \{0, 1, 2, \dots, d\}, i \in \{1, \dots, \ell\}, \ell \in \mathbb{N}\right\} \cup \{v\},$$

where $v$ denotes the multi-index of length zero. Given $\alpha$ and $\beta$, we write $\alpha \star \beta$ for the concatenation of the indices. For instance, $(0, 1) \star (1, 0, 0) = (0, 1, 1, 0, 0)$. Lastly, we write $\alpha-$ for the multi-index in $\mathcal{M}$ obtained by deleting the last component of $\alpha$. For instance, $(1, 0, 0)- = (1, 0)$.

**Definition 5.12** Write $(X_t) = (t, W_t)$ for time-augmented Brownian motion. For any $\alpha \in \mathcal{M}$, we define the iterated integrals of time-augmented Brownian motion

$$I_{\alpha;t} := \begin{cases} 1, & \text{if } \ell(\alpha) = 0, \\ \int_0^t I_{\alpha-;s}\,ds, & \text{if } \ell \geq 1 \text{ and } j_\ell = 0, \\ \int_0^t I_{\alpha-;s}\,dW_s^{j_\ell}, & \text{if } \ell \geq 1 \text{ and } j_\ell \neq 0. \end{cases}$$

We call linear combinations of these integrals *signature payoffs*.

In contrast to simple payoffs, signature payoffs are only defined almost surely on $C^0([0, T]; \mathbb{R}^d)$. One can define a signature payoff surely if the lift $\mathbb{W}$ is given in addition to the trace.

**Lemma 5.13** *Suppose that $\alpha \in \mathcal{M}$. If $\ell(\alpha) > n(\alpha)$, we can write $\alpha$ uniquely as $\beta \star (i) \star (0)^k$ for some $k \in \{0, 1, 2, \dots\}$, $\beta \in \mathcal{M}$ and $1 \leq i \leq d$. Furthermore,*

$$I_{\alpha;T} = \int_0^T \frac{(T-t)^k}{k!} I_{\beta;t}\,dW_t^i.$$

*This holds almost surely if $W$ is taken to be a Brownian motion and the integrals are Itô integrals, and surely if $W$ is taken to be a rough path with a given lift $\mathbb{W}$ and all integrals are taken to be rough-path integrals.*

***Proof*** We proceed by induction on $k$. The case $k = 0$ is true by the definition of the iterated integrals. From the induction hypothesis and either the stochastic Fubini theorem (see Protter [40, Chap. V.4]) or the rough-path Fubini theorem (see Gerasimovičs and Hairer [23]), we may compute

$$\begin{aligned} I_{\beta\star(i)\star(0)^{k+1};T} &= \int_0^T I_{\beta\star(i)\star(0)^k;t}\,dt \\ &= \int_0^T \int_0^t \frac{(t-s)^k}{k!} I_{\beta;s}\,dW_s^i\,dt \\ &= \int_0^T \int_s^T \frac{(t-s)^k}{k!} I_{\beta;s}\,dt\,dW_s^i \\ &= \int_0^T \frac{(T-s)^{k+1}}{(k+1)!} I_{\beta;s}\,dW_s^i. \end{aligned}$$

□

**Theorem 5.14** *For any $\alpha \in \mathcal{M}$, the payoff $I_{\alpha;T}$ is $W$-controllable. Furthermore, we may compute the delta and gamma in the case of the Bachelier model $\tilde{S} = W$ using for $1 \le i_1, i_2 \le d$ the formulae*

$$(\Delta_t^{I_{\alpha;T}})^{i_2} = \begin{cases} 0, & \text{if } \ell(\alpha) = n(\alpha), \\ \frac{(T-t)^k}{k!} I_{\gamma\star(0);t}, & \text{if } \alpha = \gamma\star(0,i_2)\star(0)^k, \\ \frac{(T-t)^k}{k!} I_{\gamma\star(i_1);t}, & \text{if } \alpha = \gamma\star(i_1,i_2)\star(0)^k, \\ \frac{(T-t)^k}{k!}, & \text{if } \alpha = (i_2)\star(0)^k, \\ 0, & \text{otherwise} \end{cases}$$

*and*

$$(\Gamma_t^{I_{\alpha;T}})^{i_1,i_2} = \begin{cases} 0, & \text{if } \ell(\alpha) = n(\alpha), \\ 0, & \text{if } \alpha = \gamma\star(0,i_2)\star(0)^k, \\ \frac{(T-t)^k}{k!} I_{\gamma;t}, & \text{if } \alpha = \gamma\star(i_1,i_2)\star(0)^k, \\ 0, & \text{otherwise.} \end{cases}$$

***Proof*** Whenever $\ell(\alpha) = n(\alpha)$, $I_{\alpha;T}$ is a deterministic function of $T$ (or a constant) so that $(\Delta^{I_{\alpha;T}}, \Gamma^{I_{\alpha;T}}) \equiv (0,0)$. Suppose now that $\ell(\alpha) > n(\alpha)$ and $\alpha = \beta\star(i_2)\star(0)^k$ for some $k \in \{0,1,2,\dots\}$, $\beta \in \mathcal{M}$ and $1 \le i_2 \le d$. Then by the definition of the delta and Lemma 5.13, we immediately deduce that $\Delta_t^{I_{\alpha;T}} = \frac{(T-t)^k}{k!} I_{\beta;t}$. For the gamma, if $\beta = \gamma\star(0)$, then $I_{\beta;t}$ is a Young integral and thus has vanishing Gubinelli derivative. If $\beta = \gamma\star(i_1)$, then the definition of $I_{\beta;t}$ and properties of the rough-path integral give

$$\left(\int_0^t I_{\gamma\star(i_1);u}\,dW_u\right)' = I_{\gamma;t}.$$

By the Leibniz rule of Gubinelli derivatives, we deduce the result. □

The so-called universal approximation theorem (see Lyons et al. [34, Proposition 4.5] shows that derivative payoffs which are continuous in an appropriate topology (see the reference for details) can be approximated with arbitrary accuracy in the sup-norm by a finite linear combination of these iterated integrals on any compact set of rough paths. If one has a probability model on the space of paths, one can choose the compact set to ensure that the probability of a path not lying in this set is arbitrarily small. Such a linear combination of iterated integrals is called a linear signature payoff function. We have shown that all linear signature payoff functions are $W$-controllable.

However, this is not an entirely satisfactory result from a financial point of view as the signature payoffs are only defined almost surely, as are their delta and gamma. As a result, signature payoffs are not continuously $W$-controllable. However, if one supposes that the iterated integrals defining the delta and gamma are known at all times, then Lemma 5.13 shows that (3.1) still holds surely.

In contrast to simple payoffs, the gamma of a signature payoff need not be a symmetric tensor. This implies that general signature payoffs cannot be replicated using the gamma-hedging strategy using just European options.

### 5.2.4 Barrier options

***Example 5.15*** The time-0 price of the no-touch option with maturity $T$ and barrier $B > 0 = W_0$ in the Bachelier model with $\tilde{S} = W$ is $X_0^B = 2\Phi(\frac{B}{\sqrt{T}}) - 1$. Consequently, the price, delta and gamma of the option at some time $t \in [0, T]$ are respectively given by

$$X_t^B = \mathbb{1}_{\{M_t < B\}}\left(2\Phi\left(\frac{B - W_t}{\sqrt{T-t}}\right) - 1\right),$$

$$\Delta_t^B = \frac{2\mathbb{1}_{\{M_t < B\}}}{\sqrt{T-t}}\phi\left(\frac{B - W_t}{\sqrt{T-t}}\right),$$

$$\Gamma_t^B = \frac{2(B - W_t)\mathbb{1}_{\{M_t < B\}}}{(T-t)^{\frac{3}{2}}}\phi\left(\frac{B - W_t}{\sqrt{T-t}}\right).$$

Here $\Phi$ is the distribution function of the standard normal, $\phi$ its density and $M$ is the one-sided supremum process of $W$. In particular, this option is not $W$-controllable because both the delta and gamma may have discontinuous sample paths.

In order to obtain a controllable payoff, we need to smooth the barrier option in some way. The benefits of smoothing payoffs are well known to practitioners; for example, smoothing of the payoff function is often done to obtain better convergence of numerical schemes, even for vanilla options; see for example Heston and Zhou [28] and Wade et al. [44].

**Definition 5.16** A smooth function $\rho : \mathbb{R}^n \to \mathbb{R}$ is said to be a *mollifier* if

1) $\rho$ is compactly supported;
2) $\rho$ is zero outside of a compact subset of the open unit ball;
3) $\int_{\mathbb{R}^n} \rho(x)\, dx = 1$.

**Definition 5.17** Let $g : [0,T] \to \mathbb{R}$ and $W$ be a Brownian motion. We denote by $W^{-g}$ *the Brownian motion* $W$ *drifted by the function* $g$. That is, $W_t^{-g} := W_t - g(t)$ for each $t \in [0,T]$. We denote the corresponding *running maximum* and *running minimum* by $M^{-g}$ and $m^{-g}$ respectively. That is, $M_t^{-g} := \sup_{s\leq t} W_s^{-g}$ and $m_t^{-g} := \inf_{s\leq t} W_s^{-g}$ for each $t \in [0,T]$.

**Definition 5.18** Given $\epsilon_1 < 0 < \epsilon_2$, smooth functions $g_1, g_2 : [0,T] \to \mathbb{R}$ satisfying $g_1(t) < g_2(t)$ for $t \in [0,T]$ and another smooth function $f : \mathbb{R} \to [0,1]$ such that

$$f(x) = \begin{cases} 1, & \text{if } g_1(T) < x < g_2(T), \\ 0, & \text{if } x \leq g_1(T) + \frac{1}{2}\epsilon_1 \text{ or } g_2(T) + \frac{1}{2}\epsilon_2 \leq x, \end{cases}$$

we define a payoff function by

$$F^{\epsilon_1,\epsilon_2}(g_1,g_2;f)(W_\cdot) := \mathbb{1}_{\{\epsilon_1 < m_T^{-g_1}, M_T^{-g_2} < \epsilon_2\}} f(W_T).$$

Given in addition a mollifier $\rho$ and $b_1, b_2 \in (0,\infty)$ we define the payoff of the *smoothed no-touch* to be

$$\begin{aligned} &G^{b_1,b_2,\rho}(\epsilon_1,\epsilon_2,g_1,g_2;f) \\ &:= \int_{\mathbb{R}^2} \frac{1}{b_1 b_2} \rho\left(\frac{x_1}{b_1}, \frac{x_2}{b_2}\right) F^{\epsilon_1,\epsilon_2}(g_1 - x_1, g_2 - x_2; f)\, dx_2\, dx_1. \end{aligned}$$

**Theorem 5.19** *The smoothed no-touch payoff* $G = G^{b_1,b_2,\rho}(\epsilon_1,\epsilon_2,g_1,g_2;f)$ *is continuously* $W$*-controllable for sufficiently small positive values of* $b_1$ *and* $b_2$.

***Proof*** See the Appendix. □

### 5.3 Superhedging

Simple payoffs are dense in $L^2$, and the universal approximation theorem allows one to approximate derivatives using signature payoffs outside a set of low probability. Neither result is particularly satisfactory as we should like to know which derivatives can be approximated surely. In addition, we should like to know which options can be approximated by replication strategies using only European options or other realistic financial derivatives.

For simplicity, we only consider the case of 1-dimensional noise. We take $\Omega$ to be the space of continuous $\mathbb{R}$-valued functions on $[0,T]$ which start at 0, equipped with the sup-norm.

**Definition 5.20** A derivative with continuous payoff $F : \Omega \to \mathbb{R}$ is said to be *approximately controllable* if there exists a family of continuously $W$-controllable derivatives with payoffs $(G^\epsilon)_{\epsilon>0}$ such that

1) $G^\epsilon(\omega) \geq F(\omega)$ for all $\epsilon > 0$ and $\omega \in \Omega$;
2) $\lim_{\epsilon\to 0} \mathbb{E}[G^\epsilon] = \mathbb{E}[F]$.

Such a family $(G^\epsilon)_{\epsilon>0}$ of payoffs is said to be a *family of controllable payoffs approximating* $F$.

**Theorem 5.21** *Let $F : \Omega \to \mathbb{R}$ be continuous with finite expectation. Suppose there exists a continuously $W$-controllable $G : \Omega \to \mathbb{R}$ with $G(\omega) \geq F(\omega)$ for all $\omega \in \Omega$ and with $\mathbb{E}[G] < \infty$. Then $F$ is approximately controllable.*

***Proof*** Without loss of generality, we may assume that $G \equiv 0$. The general case then follows by applying the proof to $G - F$. Given $N \in \mathbb{N}$, define

$$F_N(\omega) := \frac{1}{N} - \frac{1}{N}\sum_{n=1}^{N^2} \mathbb{1}_{\{\omega \in F^{-1}(-\infty, -\frac{n}{N})\}}.$$

We may express $F_N$ in terms of the ceiling function as

$$\begin{aligned} F_N(\omega) &= \frac{1}{N} - \frac{1}{N}\sum_{n=1}^{N^2} \mathbb{1}_{\{n < -NF(\omega)\}} \\ &= \frac{1}{N} + \frac{1}{N}\max\left\{ -\sum_{n=1}^{\infty} \mathbb{1}_{\{n<-NF(\omega)\}}, -N^2 \right\} \\ &= \frac{1}{N} + \max\left\{ -\frac{\lceil -NF(\omega)\rceil}{N}, -N \right\}. \end{aligned}$$

We now see that $F_N(\omega) \geq \frac{1}{N} - \frac{\lceil -NF(\omega)\rceil}{N} \geq F(\omega)$ and $F_N(\omega) \to F(\omega)$ as $N \to \infty$. By the continuity of $F$, $F^{-1}(-\infty, -\frac{n}{N})$ is open and so $F_N$ is measurable. By the dominated convergence theorem, $\lim_{N\to\infty} \mathbb{E}[F_N] = \mathbb{E}[F]$, using $|F| + 1$ as the dominating function for $F_N$. So for any $\epsilon > 0$, we may fix $N$ large enough to ensure that

$$\mathbb{E}[F_N] < \mathbb{E}[F] + \frac{\epsilon}{4}.$$

As $F^{-1}(-\infty, -\frac{n}{N})$ is an open set in a separable space, we may write it as a countable union of open balls in $\Omega$. More precisely, let $B(h, \epsilon)$ denote the open ball around $h \in C^\infty([0, T]; \mathbb{R}) \cap \Omega$ in the sup-norm. That is,

$$\omega \in B(h, \epsilon) \iff \|\omega - h\|_{\infty;[0,T]} < \epsilon \iff -\epsilon < m_T^{-h}(\omega), M_T^{-h}(\omega) < \epsilon.$$

As smooth functions are dense in $\Omega$, we have for each $n \in \{1, \ldots, N^2\}$ that there exist $(h_{i,n})_{i\geq 1} \in C^\infty$ and positive $(\epsilon_{i,n})_{i\geq 1}$ such that

$$F^{-1}\left(-\infty, -\frac{n}{N}\right) = \bigcup_{i=1}^{\infty} B(h_{i,n}, \epsilon_{i,n}).$$

We can choose $M_n \in \mathbb{N}$ sufficiently large so that

$$\mathbb{Q}\left[F^{-1}\left(-\infty, -\frac{n}{N}\right)\right] - \frac{\epsilon}{4N} \leq \mathbb{Q}\left[\bigcup_{i=1}^{M_n} B(h_{i,n}, \epsilon_{i,n})\right] \leq \mathbb{Q}\left[F^{-1}\left(-\infty, -\frac{n}{N}\right)\right].$$

Define

$$\tilde{F} := \frac{1}{N} - \frac{1}{N}\sum_{n=1}^{N^2} \mathbb{1}_{\bigcup_{i=1}^{M_n} B(h_{i,n},\epsilon_{i,n})}.$$

Then

$$\mathbb{E}[\tilde{F}] < \mathbb{E}[F_N] + \frac{\epsilon}{4} < \mathbb{E}[F] + \frac{\epsilon}{2},$$
$$F(\omega) \leq F_N(\omega) \leq \tilde{F}(\omega), \qquad \forall \omega \in \Omega. \tag{5.4}$$

For each $n \in \{1, \ldots, N^2\}$ and $i \in \{1, \ldots, M_n\}$, define

$$F_{i,n}(\omega) := \begin{cases} 1, & \{\|\omega - h_{i,n}\|_{\infty;[0,T]} > \epsilon_{i,n}\}, \\ 0, & \text{otherwise.} \end{cases}$$

For each $n \in \{1, \ldots, N^2\}$, let

$$\tilde{F}_n(\omega) := \prod_{i=1}^{M_n} F_{i,n}(\omega) = \begin{cases} 1, & \text{if } \omega \in \bigcap_{i=1}^{M_n} B(h_{i,n}, \epsilon_{i,n})^c, \\ 0, & \text{otherwise.} \end{cases}$$

So $1 - \tilde{F}_n(\omega) = \mathbb{1}_{\bigcup_{i=1}^{M_n} B(h_{i,n},\epsilon_{i,n})}$ and by the inclusion–exclusion principle,

$$\mathbb{E}[1 - \tilde{F}_n] = \mathbb{Q}\left[\bigcup_{i=1}^{M_n} B(h_{i,n}, \epsilon_{i,n})\right] = \sum_{k=1}^{M_n}\left((-1)^{k-1} \sum_{\substack{I \subseteq \{1,\ldots,M_n\} \\ |I|=k}} \mathbb{Q}[B_I]\right),$$

where $B_I = \bigcap_{i\in I} B(h_{i,n}, \epsilon_{i,n})$. For any $I \subseteq \{1, \ldots, M_n\}$, the payoff $\mathbb{1}_{B_I}$ is a no-touch option with the time-dependent barriers

$$t \mapsto \pm \inf_{i\in I}\left(\epsilon_{i,n} + h_{i,n}(t)\right).$$

These barriers are in general no longer smooth, but they are continuous on the bounded interval $[0, T]$. Thus they may be approximated by smooth functions from below/above with arbitrary accuracy. This means that we can approximate the payoffs $\tilde{F}_n$ using a family of payoffs $\tilde{G}_n^\delta$ such that

1) $\tilde{F}_n \leq \tilde{G}_n^\delta$ for each $\delta > 0$;
2) $\lim_{\delta\to 0} \mathbb{E}[\tilde{G}_n^\delta - \tilde{F}_n] = 0$;
3) $\tilde{G}_n^\delta$ is a finite sum of no-touch options with smooth time-dependent barriers for each $\delta > 0$.

By Theorem 5.19, we can approximate each $\tilde{G}_n^\delta$ with a family of smoothed no-touch options, which are continuously $W$-controllable payoffs. These payoffs $(\tilde{G}_n^{\delta,\rho})_\rho$ satisfy $\tilde{G}_n^{\delta,\rho} \geq \tilde{G}_n^\delta$ and $\lim_{\rho\to 0} \mathbb{E}[\tilde{G}_n^{\delta,\rho}] = \mathbb{E}[\tilde{G}_n^\delta]$. By construction, $\tilde{G}_n^{\delta,\rho} \geq \tilde{G}_n^\delta \geq \tilde{F}_n$ and

$$\lim_{\rho\to 0}\lim_{\delta\to 0} \mathbb{E}[\tilde{G}_n^{\delta,\rho} - \tilde{F}_n] = \lim_{\rho\to 0} \mathbb{E}[\tilde{G}_n^{\delta,\rho} - \tilde{G}_n^\delta] + \lim_{\delta\to 0} \mathbb{E}[\tilde{G}_n^\delta - \tilde{F}_n] = 0.$$

So we can pick some $\rho$ and $\delta > 0$ sufficiently small so that $\mathbb{E}[\tilde{G}_n^{\delta,\rho} - \tilde{F}_n] \leq \frac{\epsilon}{2N}$ for each $n \in \{1, \ldots, N^2\}$. Finally, define the $W$-controllable payoff

$$\bar{G}^\epsilon := \frac{1}{N} - \frac{1}{N}\sum_{n=1}^{N^2}(1 - \tilde{G}_n^{\delta,\rho}).$$

Then

$$\begin{aligned}
\mathbb{E}[\bar{G}^\epsilon] &= \frac{1}{N} - \frac{1}{N}\sum_{n=1}^{N^2}\mathbb{E}[1 - \tilde{G}_n^{\delta,\rho}] \\
&\leq \frac{1}{N} - \frac{1}{N}\sum_{i=1}^{N^2}(\mathbb{E}[1 - \tilde{F}_n] - \mathbb{E}[\tilde{G}_n^{\delta,\rho} - \tilde{F}_n]) \\
&\leq \frac{1}{N} - \frac{1}{N}\sum_{n=1}^{N^2}\left(\mathbb{Q}\left[\bigcup_{i=1}^{M_n} B(h_{i,n}, \epsilon_{i,n})\right] - \frac{\epsilon}{2N}\right) \\
&= \left(\frac{1}{N} - \frac{1}{N}\sum_{n=1}^{N^2}\mathbb{Q}\left[\bigcup_{i=1}^{M_n} B(h_{i,n}, \epsilon_{i,n})\right]\right) + \frac{\epsilon}{2} \\
&= \mathbb{E}[\tilde{F}] + \frac{\epsilon}{2} \\
&< \mathbb{E}[F] + \epsilon.
\end{aligned}$$

We also see that

$$\begin{aligned}
\bar{G}^\epsilon &= \frac{1}{N} - \frac{1}{N}\sum_{n=1}^{N^2}\left((1 - \tilde{F}_n) + (\tilde{F}_n - \tilde{G}_n^{\delta,\rho})\right) \\
&\geq \frac{1}{N} - \frac{1}{N}\sum_{i=1}^{N^2}\mathbb{1}_{\bigcup_{i=1}^{M_n} B(h_{i,n},\epsilon_{i,n})} = \tilde{F} \geq F,
\end{aligned}$$

using (5.4). It follows that for each $\epsilon > 0$, there exists a family of controllable payoffs, namely $(\bar{G}^\epsilon)_{\epsilon>0}$, which approximate $F$ as required. □

To extend our result to higher-dimensional noise, the additional ingredient needed is to prove that higher-dimensional no-touch options with smooth barrier manifolds can be superhedged by using $W$-controllable payoffs. This requires a higher-dimensional analogue of Theorem 5.19, which can be proved in the same way using PDE theory. In this case, the gamma of the smoothed option can be related to the partial derivatives of the solution to the PDE in a formula analogous to (A.5) in the Appendix. Hence in higher dimensions, one can superhedge continuous payoffs $F$ with $W$-controllable hedging instruments that have symmetric gamma tensors such as European options. By contrast, signature payoffs representing Lévy-area terms need not be continuous and can have non-symmetric gamma.

## 6 Conclusion

We have shown how the gamma-hedging strategy may provide some insight into the benefits of the market practices of calibrating pricing models to market prices and of gamma hedging.

The central ingredient in our approach is to develop rough-path versions of the Clark–Ocone formula, and we have given a number of examples. The thesis [31] studies the relationship between Malliavin calculus and the Gubinelli derivative, defining a closable operator for whose domain these notions coincide and hence for which a Clark–Ocone formula holds. The same thesis also explores the relationship between the Gubinelli derivative and functional Itô calculus.

## Appendix: Proofs

***Proof of Theorem 2.3*** The result is proved in the case of Hölder norms in Friz and Hairer [21, Theorem 4.10]. We show only the changes needed for $p$-variation results. Suppose $s \leq u \leq t$. Set

$$\Xi_{s,t} = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t};$$

then we have

$$\delta\Xi_{s,u,t} := \Xi_{s,t} - \Xi_{s,u} - \Xi_{u,t} = -R^Y_{s,u} X_{u,t} - Y'_{s,u} \mathbb{X}_{u,t}.$$

Since $p < 3$, $x \to x^{\frac{p}{3}}$ is concave so that

$$\begin{aligned} |\delta\Xi_{s,u,t}|^{\frac{p}{3}} &\leq \|R^Y\|^{\frac{p}{3}}_{\frac{p}{2}\text{-var};[s,u]} \|X\|^{\frac{p}{3}}_{p\text{-var};[u,t]} + \|Y'\|^{\frac{p}{3}}_{p\text{-var};[s,u]} \|\mathbb{X}\|^{\frac{p}{3}}_{\frac{p}{2}\text{-var};[u,t]} \\ &= w_{R^Y,\frac{p}{2}}(s,t)^{\frac{2}{3}} w_{X,p}(s,t)^{\frac{1}{3}} + w_{Y',p}(s,t)^{\frac{1}{3}} w_{\mathbb{X},\frac{p}{2}}(s,t)^{\frac{2}{3}}. \end{aligned}$$

Applying Lemma 2.2, we see that the right-hand side is a control function. The existence of the limit and the bound now follow from Armstrong et al. [4, Proposition 3.5]. The argument of [21, Theorem 4.10] establishes the continuity of the integral map. □

***Proof of Lemma 4.2*** The Black–Scholes price satisfies

$$\mathrm{BS}^f(s_1, s_2, \tau) = \int_0^\infty f(u) q(s_1, s_2, \tau, u)\, du, \tag{A.1}$$

where the pricing kernel $q$ is given by

$$q(s_1, s_2, \tau, u) = \frac{1}{u s_2 \sqrt{2\pi\tau}} \exp\left( - \frac{(\log u - \log s_1 + \frac{1}{2} s_2^2 \tau)^2}{2 s_2^2 \tau} \right).$$

A direct computation shows that

$$(s_1^2 s_2 \tau)\frac{\partial^2}{\partial s_1^2} q(s_1, s_2, \tau, u) = \frac{\partial}{\partial s_2} q(s_1, s_2, \tau, u),$$

which combined with (A.1) gives the result. □

**Lemma A.1** *Let $f : (0, \infty) \to \mathbb{R}$ be a convex, nonlinear payoff function. Then its pricing function $\mathrm{BS}^f(s_1, s_2, \tau)$ satisfies $\frac{\partial^2 BS^f}{\partial s_1^2} > 0$ for all $\tau > 0$.*

***Proof*** Using the explicit formula (A.1) for the pricing kernel in the Black–Scholes model and making the substitution $v = \frac{u}{s_1}$, we may write

$$\mathrm{BS}^f(s_1, s_2, \tau) = \int_0^\infty f(s_1 v)\frac{1}{v s_2 \sqrt{2\pi\tau}} \exp\left( - \frac{(\log v + \frac{1}{2}s_2^2\tau)^2}{2 s_2^2 \tau}\right) dv. \tag{A.2}$$

Thus as a function of $s_1$, $\mathrm{BS}^f(s_1, s_2, \tau)$ is a convex combination of the convex functions $s_1 \to f(s_1 v)$; so it must itself be convex. Since $\mathrm{BS}^f$ is smooth for $\tau > 0$, we conclude that $\frac{\partial^2 \mathrm{BS}^f}{\partial s_1^2} \geq 0$.

If $f$ were in addition smooth and had a positive second derivative at some point $x$, then by considering the integral (A.2) in a neighborhood of $v = \frac{x}{s_1}$, we could conclude that $\frac{\partial^2 \mathrm{BS}^f}{\partial s_1^2}$ is strictly positive for any $\tau > 0$. On the other hand, if $f'' \equiv 0$ then $\frac{\partial^2 \mathrm{BS}^f}{\partial s_1^2} \equiv 0$.

Since we can think of a derivative with maturity $T$ in the Black–Scholes model as being a derivative with maturity $T - \epsilon$ and payoff $\mathrm{BS}^f(s_1, s_2, \tau - \epsilon)$, we can combine the observations of the last two paragraphs to see that $\frac{\partial^2 \mathrm{BS}^f}{\partial s_1^2}$ is either identically zero for $\tau > \epsilon$ or strictly positive for $\tau > \epsilon$. Hence either $\frac{\partial^2 \mathrm{BS}^f}{\partial s_1^2} \equiv 0$ for $\tau > 0$, or $\frac{\partial^2 \mathrm{BS}^f}{\partial s_1^2} > 0$ for all $\tau > 0$. Suppose for contradiction the former occurs. It then follows from the Black–Scholes PDE that $\frac{\partial \mathrm{BS}^f}{\partial \tau} = 0$. Since $\mathrm{BS}^f$ is linear in $s_1$ for $\tau > 0$, by the continuity of $\mathrm{BS}^f$, $f$ must be linear too, which is a contradiction. Hence $\frac{\partial^2 \mathrm{BS}^f}{\partial s_1^2} > 0$ for all $\tau < 0$. □

***Proof of Proposition 5.2*** Using [21, Proposition 7.1], the right-hand side rough-path integral is equal to

$$\big((Y, Y') \cdot (M, \mathbb{M}^{\mathrm{Itô}})\big)_{0,t} = \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} (Y_u M_{u,v} + Y'_u \phi'_u \mathbb{W}^{\mathrm{Itô}}_{u,v}).$$

For any continuous adapted process $Y$, the Itô integral against a local martingale $M$ is defined as the limit in probability

$$\int_0^t Y_u \, dM_u := \mathbb{P}\text{-}\lim_{n \to \infty} \sum_{[u,v] \in \pi_n} Y_u M_{u,v},$$

where $(\pi_n)_{n\in\mathbb{N}}$ is any refining sequence of finite partitions with $\lim_{n\to\infty}|\pi_n| = 0$. By switching to a subsequence, we can assume that the convergence holds almost surely. Following the proof of [21, Proposition 5.1], it now suffices to show that the difference $\sum_{[u,v]\in\pi_n} Y'_u\phi'_u\mathbb{W}^{\text{Itô}}_{u,v}$ between the Itô and the rough-path integrals vanishes on a set of full measure as $n\to\infty$. It suffices to show that

$$\left\| \sum_{[u,v]\in\pi} Y'_u\phi'_u\mathbb{W}^{\text{Itô}}_{u,v} \right\|^2_{L^2} = \mathcal{O}(|\pi|),$$

as convergence in $L^2$ implies convergence in probability, and convergence in probability implies almost sure convergence along a subsequence. For a given finite partition $\pi\in\mathcal{P}[0,T]$, define the piecewise constant martingale $N$ with $N_0 := 0$ which is constant on the intervals $[t_k, t_{k+1})\in\pi$ and with increments

$$N_{t_{k+1}} - N_{t_k} = Y'_{t_k}\phi'_{t_k}\mathbb{W}^{\text{Itô}}_{t_k,t_{k+1}}.$$

Define also the stopping times $\tau_K := \inf\{t\in[0,T] : \|Y'_t\phi'_t\|_{\max}\geq K\}$ for each $K\in\mathbb{N}$, where $\|\cdot\|_{\max}$ denotes the max norm for matrices (which picks the element with the largest absolute value). The stopped process $N^{\tau_K}$ is also a martingale and for each $K\in\mathbb{N}$, we have

$$\begin{aligned}
\left\| \sum_{[u,v]\in\pi} Y'_{u\wedge\tau_K}\phi'_{u\wedge\tau_K}\mathbb{W}^{\text{Itô}}_{u,v} \right\|^2_{L^2} &= \left\| \sum_{k=0}^{n-1}(N^{\tau_K}_{t_{k+1}} - N^{\tau_K}_{t_k}) \right\|^2_{L^2} \\
&= \sum_{k=0}^{n-1}\|N^{\tau_K}_{t_{k+1}} - N^{\tau_K}_{t_k}\|^2_{L^2} \\
&\leq K^2\sum_{k=0}^{n-1}\|\mathbb{W}^{\text{Itô}}_{t_k,t_{k+1}}\|^2_{L^2} \\
&= \mathcal{O}(|\pi|),
\end{aligned}$$

where we have used that $\|\mathbb{W}^{\text{Itô}}_{t_k,t_{k+1}}\|^2_{L^2}$ is proportional to $|t_{k+1}-t_k|^2$ in the last line. As this is true for all $K\in\mathbb{N}$ and $\tau_K\to\infty$ as $K\to\infty$, this concludes the proof. □

To prove the equivalence of the notions of $\tilde{S}$- and $W$-controllable, we use a sequence of lemmas.

**Lemma A.2** *If $\sigma(s,t)$ invertible for all $s\in U$, $t\in[0,T]$ and if $\mu = 0$, then $G$ is $\tilde{S}$-controllable if and only if there exist adapted processes $Y$, $Y'$ such that for any $0\leq s<t\leq T$,*

$$\mathbb{E}_t[G] - \mathbb{E}_s[G] = \left((Y,Y')\cdot(\tilde{S},\tilde{\mathbb{S}})\right)_{s,t} \qquad \textit{almost surely.}$$

*Moreover, $Y$ and $Y'$ are unique up to indistinguishability and $Y = \Delta^G$.*

***Proof*** Suppose $G$ is $\tilde{S}$-controllable. Equation (5.1) establishes the Itô integral formula

$$\mathbb{E}_t[G] - \mathbb{E}_s[G] = \int_s^t \Delta_u \, d\tilde{S}_u.$$

By our assumption on $G$, $\Delta$ almost surely admits a Gubinelli derivative $\Gamma$. We may now apply Proposition 5.2 to obtain the "only if" part of our claim.

Suppose that a suitable $(Y, Y')$ exists. By the definition of $\Delta^G$ in (5.1) and the uniqueness in the martingale representation theorem, $Y$ must equal $\Delta^G$. Our result then amounts to showing uniqueness of the Gubinelli derivative up to indistinguishability of processes. According to [21, Proposition 6.4], if a rough path has the property of being "truly rough", then the Gubinelli derivative is unique. According to [21, Theorem 6.6], Brownian motion is truly rough for the values of $p$ we consider. Using Proposition 5.2 and the fact that $W$ is a solution to the stochastic differential equation

$$dW_t = \sigma^{-1}(\tilde{S}_t, t)\, d\tilde{S}_t, \qquad W_0 = 0,$$

we see that the Gubinelli derivative with respect to $\tilde{S}$ must also be unique. □

**Lemma A.3** *Assume $\sigma(s,t)$ is invertible for all $s \in U$ and $t \in [0,T]$ and that $\mu = 0$. Let $\tilde{S}(\omega)$ denote a path from the diffusion (1.2) such that $\tilde{S} \in C^{p\text{-var}}([0,T];\mathbb{R}^d)$ and the underlying Brownian path $W \in C^{p\text{-var}}([0,T];\mathbb{R}^d)$. Then*

$$(Z, Z') \in \mathcal{D}_W^{p\text{-var}} \qquad \Longrightarrow \qquad \Big(Z, Z'\big(\sigma(\tilde{S}, \cdot)\big)^{-1}\Big) \in \mathcal{D}_{\tilde{S}}^{p\text{-var}}.$$

***Proof*** By Proposition 5.2, $(\tilde{S}, \sigma(\tilde{S}, \cdot)) \in \mathcal{D}_W^{p\text{-var}}([0,T];\mathbb{R}^d)$. By the hypothesis, we know that the remainder $R^S$ given by $R^S_{s,t} := \tilde{S}_{s,t} - \sigma(\tilde{S}_s, s)W_{s,t}$ is of finite $\frac{p}{2}$-variation. Let $R^W_{s,t} := W_{s,t} - (\sigma(\tilde{S}_s, s))^{-1}\tilde{S}_{s,t}$. Thus

$$R^S_{s,t} = \tilde{S}_{s,t} - \sigma(\tilde{S}_s, s)\Big(R^W_{s,t} + \big(\sigma(\tilde{S}_s, s)\big)^{-1}\tilde{S}_{s,t}\Big) = -\sigma(\tilde{S}_s, s)R^W_{s,t}.$$

Hence $R^W$ also defines a remainder of finite $\frac{p}{2}$-variation so that

$$\Big(W, \big(\sigma(\tilde{S}, \cdot)\big)^{-1}\Big) \in \mathcal{D}_{\tilde{S}}^{p\text{-var}}([0,T];\mathbb{R}^d).$$

By the change of signal formula in [21, Proposition 7.1], it follows that

$$\Big(Z, Z'\big(\sigma(\tilde{S}, \cdot)\big)^{-1}\Big) \in \mathcal{D}_{\tilde{S}}^{p\text{-var}}.$$ □

**Lemma A.4** *Let $(\tilde{S}_t)_{t\in[0,T]}$ be a diffusion of the form (1.2) with invertible $\sigma$ and $\mu = 0$. Then it holds almost surely that*

$$(Y, Y') \in \mathcal{D}_{\tilde{S}}^{p\text{-var}}([0,T];\mathbb{R}^d) \qquad \Longrightarrow$$
$$\Big(Y^\top\sigma(\tilde{S}, \cdot), Y^\top\nabla\sigma(\tilde{S}, \cdot)\sigma(\tilde{S}, \cdot) + \sigma(\tilde{S}, \cdot)(Y')^\top\Big) \in \mathcal{D}_W^{p\text{-var}}([0,T];\mathbb{R}^d).$$

*Conversely*,

$$(Y, Y') \in \mathcal{D}_W^{p\text{-var}}([0,T];\mathbb{R}^d) \quad \Longrightarrow$$
$$\left(Y^\top A, Y^\top A' + A(Y')^\top\right) \in \mathcal{D}_{\tilde{S}}^{p\text{-var}}([0,T];\mathbb{R}^d),$$

*where $A$ denotes the inverse of $\sigma(\tilde{S},\,\cdot\,)$ and $A'$ its Gubinelli derivative with respect to $\tilde{S}$. Hence $G$ is $\tilde{S}$-controllable if and only if $G$ is $W$-controllable.*

***Proof*** For the first part, we know from Proposition 5.2 that

$$\left(\tilde{S}, \sigma(\tilde{S},\,\cdot\,)\right) \in \mathcal{D}_W^{p\text{-var}}([0,T];\mathbb{R}^d)$$

so that $(Y^\top, (Y')^\top\sigma(\tilde{S},\,\cdot\,)) \in \mathcal{D}_W^{p\text{-var}}([0,T];\mathbb{R}^d)$ by the chain rule for Gubinelli derivatives. Similarly,

$$\left(\sigma(\tilde{S},\,\cdot\,), \nabla\sigma(\tilde{S},\,\cdot\,)\right) \in \mathcal{D}_{\tilde{S}}^{p\text{-var}}([0,T];\mathbb{R}^{d\times d})$$

and so $(\sigma(\tilde{S},\,\cdot\,), (\nabla\sigma(\tilde{S},\,\cdot\,))\sigma(\tilde{S},\,\cdot\,))$ belongs to $\mathcal{D}_W^{p\text{-var}}([0,T];\mathbb{R}^{d\times d})$. Applying Leibniz' rule to the path $Y^\top\sigma(\tilde{S},\,\cdot\,)$ yields the result. The second part is proved analogously by using Lemma A.3. For the final part, we just apply the first two results taking $Y \equiv \Delta^G$ and $Y_t = \mathbb{E}_t[D_t G]$, respectively. □

***Proof of Proposition 5.6*** The Stratonovich enhancement of Brownian motion is defined to be

$$\mathbb{W}_{s,t}^{\text{Strat}} = \int_s^t W_{s,u} \circ dW_u,$$

where the last integral is a Stratonovich integral. It is proved in Ledoux et al. [32, Theorem 5] that for $p \in (2,3)$,

$$\mathbb{Q}[|||\mathbf{W}^{\text{Strat}}|||_{p\text{-var};[0,1]} < \epsilon] > 0, \qquad \forall \epsilon > 0,$$

where $\mathbf{W}^{\text{Strat}} = (W, \mathbb{W}^{\text{Strat}})$.

Let $\mathbf{0}^B = (0, \mathbb{O})$ denote the (nontrivial) Brownian lift of the zero path. This can be thought of intuitively as a path with a trivial trace but which has nontrivial quadratic variation, matching that of Brownian motion (see [21, Sect. 1.2] for an illuminating discussion of this type of construction). Using the fact that

$$\mathbb{W}_{s,t}^{\text{Itô}} - \mathbb{O}_{s,t}^B = \mathbb{W}_{s,t}^{\text{Strat}}$$

and the scaling property of the rough-path metric, it follows that

$$\mathbb{Q}[|||\mathbf{W}^{\text{Itô}} - \mathbf{0}^{\text{Itô}}|||_{p\text{-var};[0,T]} < \epsilon] > 0.$$

Since $\mathcal{B}^p([0,T];\mathbb{R}^d)$ is defined as the closure of a space of paths with smooth trace, we may assume by the triangle inequality without loss of generality that

$\tilde{\mathbf{W}} = (\tilde{W}, \tilde{\mathbb{W}}^B)$, where $\tilde{W}$ is smooth. Consider the rough differential equation

$$dY_t = \frac{d\tilde{W}}{dt}dt + d\mathbf{W}_t^B, \quad t \in [0,T], \qquad Y_0 = 0. \tag{A.3}$$

The solution of a rough differential equation is a controlled rough path. The Gubinelli derivative of the solution is required to match the corresponding coefficient in the differential equation. The trace of the solution is determined by requiring that the equation holds as a rough-integral equation. Hence we can write down the solution of this equation in the case that the driver is $\mathbf{W} = \mathbf{0}^B$: it is the 0-controlled rough path $(\tilde{W}, I)$, where $I$ is the constant map with value given by the identity matrix.

Given a rough path $(X, \mathbb{X})$ and an $X$-controlled rough path $(Y, Y')$, we can canonically enhance $Y$ to obtain a new rough path $(Y, \mathbb{Y})$ via the Riemann sum

$$\mathbb{Y}_{s,t} = \lim_{|\pi|\to 0}\left(\sum_{\pi} Y_u \otimes Y_{u,v} + Y' \otimes Y'\mathbb{X}_{u,v}\right).$$

Thus the canonical enhancement of $(\tilde{W}, I)$ is $\tilde{\mathbf{W}}^B$. Write $\mathbf{Y}$ for the random rough-path solution of (A.3) when the driver is $(W, \mathbb{W}^{\text{Itô}})$. Given $\epsilon > 0$, by the continuity of the solution operator for RDEs and the continuity of the canonical enhancement, we can find $\delta > 0$ such that

$$|||\mathbf{W}^{\text{Itô}} - \mathbf{0}^B|||_{p\text{-var};[0,T]} < \delta \qquad \Longrightarrow \qquad |||\mathbf{Y} - \tilde{\mathbf{W}}|||_{p\text{-var};[0,T]} < \epsilon. \tag{A.4}$$

Define a process $X$ by the Itô integral

$$X_t = -\int_0^t \left.\frac{d\tilde{W}}{dt}\right|_u dW_u.$$

Using Girsanov's theorem and Proposition 5.2, we may define an equivalent probability measure $\tilde{\mathbb{Q}}$ by the condition

$$\left.\frac{d\tilde{\mathbb{Q}}}{d\mathbb{Q}}\right|_{\mathcal{F}_t} = \exp\left(X_t - \frac{1}{2}[X]_t\right)$$

such that $\mathbf{Y}$ is the Itô enhancement of a $\tilde{\mathbb{Q}}$-Brownian motion.

We have already shown that

$$\mathbb{Q}[|||\mathbf{W}^{\text{Itô}} - \mathbf{0}^{\text{B}}|||_{p\text{-var};[0,T]} < \delta] > 0.$$

Hence by the equivalence of measures,

$$\tilde{\mathbb{Q}}[|||\mathbf{W}^{\text{Itô}} - \mathbf{0}^{\text{B}}|||_{p\text{-var};[0,T]} < \delta] > 0.$$

It now follows from (A.4) that

$$\tilde{\mathbb{Q}}[|||\mathbf{Y} - \tilde{\mathbf{W}}|||_{p\text{-var};[0,T]} < \epsilon] > 0.$$

Since $\mathbf{Y}$ is an Itô-enhanced $\tilde{\mathbb{Q}}$-Brownian motion and $\mathbf{W}^{\text{Itô}}$ is an Itô-enhanced $\mathbb{Q}$-Brownian motion, this is equivalent to the statement

$$\mathbb{Q}[|||\mathbf{W}^{\text{Itô}} - \tilde{\mathbf{W}}|||_{p\text{-var};[0,T]} < \epsilon] > 0. \qquad \square$$

To prove Proposition 5.9, we recall the following result proved by Coutin and Lejay [12, Proposition 1].

**Proposition A.5** *Let $X = (X^1, \dots, X^d)$ be an $\mathbb{R}^d$-valued continuous semimartingale and set*

$$\mathbb{X}^{Itô}_{s,t} := \int_s^t X_{s,r} \otimes dX_r := \left( \int_s^t X^i_{s,r}\, dX^j_r \right)_{1 \le i,j \le d},$$

*where the integral is in the Itô sense. Then for any $p \in (2,3)$ and almost all paths $\omega \in \Omega$, $\boldsymbol{X}(\omega) = (X(\omega), \mathbb{X}^{Itô}(\omega)) \in \mathcal{C}^{p\text{-var}}([0,T]; \mathbb{R}^d)$.*

***Proof of Proposition 5.9*** Itô integrals are special cases of continuous semimartingales; so the first statement is a direct application of Proposition A.5. The third part is a consequence of the chain rule for Gubinelli derivatives; see [21, Theorem 7.6]. For the second part, we first prove the result in the one-dimensional case $d = m = 1$. From the Dubins–Schwarz theorem, we see that for each $(s,t) \in \Delta_T$, there exists a Brownian motion $B$, possibly depending on $s$ and $t$, such that $R^X_t = R^X_s + B_{\int_s^t \phi^2_{s,u} du}$. For any $\alpha \in (\frac{1}{3}, \frac{1}{2})$, $B$ is $\alpha$-Hölder-continuous with probability one. Pick an $\omega$ so that $B(\omega)$ is $\alpha$-Hölder-continuous. We then have

$$|R^X_{s,t}(\omega)| = |B_{\int_s^t \phi^2_{s,u}\, du}(\omega)| \lesssim \left| \int_s^t \phi^2_{s,u}\, du \right|^{\alpha},$$

with the notation $x \lesssim y$ to denote an estimate of the form $x \le Cy$. We note that even though $B$ possibly depended on $(s,t)$, this dependence has now disappeared. For the purpose of easing the notation of this proof, we can assume that $\phi(\omega) \in C^{p\text{-var}}([0,T]; \mathbb{R})$. This is because $\phi$ has almost all paths of finite $p$-variation so that the subset of $\Omega$ on which both $B$ and $\phi$ are of the correct regularities is a set of full measure. As $\phi(\omega)$ is of finite $p$-variation, there exists a control function $w$ on $\Delta_T$ such that $\phi_{s,u} \le w(s,u)^{\frac{1}{p}}$ for all $(s,u) \in \Delta_T$. Thus

$$\begin{aligned} |R^X_{s,t}(\omega)|^{\frac{p}{2}} &\lesssim \left| \int_s^t w(s,u)^{\frac{2}{p}}\, du \right|^{\frac{p\alpha}{2}} \\ &\le \left( w(s,t)^{\frac{2}{p}} (t-s) \right)^{\frac{p\alpha}{2}} \\ &= w(s,t)^{\alpha} (t-s)^{\frac{p\alpha}{2}} \\ &=: \bar{w}(s,t). \end{aligned}$$

Using Lemma 2.2, we see that $\bar{w}(\cdot\,,\cdot)$ is a control function. Consequently

$$R^X(\omega) \in C^{\frac{p}{2}\text{-var}}([0,T]; \mathbb{R})$$

for almost all $\omega \in \Omega$.

Now consider the multidimensional case. For each $1 \le i \le m$ and $1 \le j \le d$, let $Y_{ij}$ denote the one-dimensional process defined by

$$(Y_{ij})_{s,t} := \int_s^t (\phi_{ij})_{s,u}\, dW_u^j$$

so that the $i$-th component of $R^X$ is given by $R_i^X = \sum_{j=1}^m Y_{ij}$. The argument of our one-dimensional proof shows that $Y_{ij} \in C^{\frac{p}{2}\text{-var}}([0,T];\mathbb{R})$ for all indices $i = 1, \dots, m$ and $j = 1, \dots, d$. By $\frac{p}{2}$-variation norm estimates of the form

$$\left\| \sum_{j=1}^m f_j \right\|_{\frac{p}{2}\text{-var}} \lesssim \sum_{j=1}^m \| f_j \|_{\frac{p}{2}\text{-var}},$$

we deduce that $R_i^X \in C^{\frac{p}{2}\text{-var}}([0,T];\mathbb{R})$ for all $i$. By the equivalence of norms in finite-dimensional spaces, we can once again use the former estimate to show that

$$R^X = (R_1^X, \dots, R_m^X) \in C^{\frac{p}{2}\text{-var}}([0,T];\mathbb{R}^m). \qquad \square$$

We prove two lemmas that allow us to prove the controllability of barrier options.

**Lemma A.6** *Consider in the Bachelier model $\tilde{S} = W$ a derivative whose payoff at maturity depends on $T$, the current level $W_T$ of the stock and $M_T$, where $M_t = \max_{s\in[0,t]} W_s$ is the running maximum of the stock. So the payoff is of the form $F = f(T, W_T, M_T)$ for some function $f : \mathbb{R}^3 \to \mathbb{R}$. Suppose that $F \in \mathbb{D}^{1,r}$. Then the delta exists and can be written as $\Delta_t^F = \mathbb{E}_t[D_t F]$ for all $t \in [0,T]$. Moreover, $(W_t, M_t)_{t\in[0,T]}$ is Markov so that we may write the price process as*

$$\mathbb{E}[F|(W_s)_{s\in[0,t]}] = \varphi(t, W_t, M_t)$$

*for some function $\varphi : [0,T] \times \mathbb{R} \times [0,\infty) \to \mathbb{R}$. If $\varphi$ is continuously differentiable in its second argument, then*

$$\Delta_t^F = \varphi_x(t, W_t, M_t)$$

*for all $t \in [0,T]$. Furthermore, if $\varphi_{xxx}$, $\varphi_{xxt}$ and $\varphi_{xxb}$ exist and are continuous, then $F$ is a $W$-controllable payoff with*

$$\Delta_t^F = \varphi_x(t, W_t, M_t) \quad \text{and} \quad \Gamma_t^F = \varphi_{xx}(t, W_t, M_t), \qquad t \in [0,T].$$

***Proof*** For a given $\omega \in \Omega$ and fixed $(u, v) \in \Delta_T$, define functions from $[0,1]$ to $\mathbb{R}$ by

$$\hat{\varphi}(\theta) := \varphi\big(u + \theta(v-u), W_u + \theta W_{u,v}, M_u + \theta M_{u,v}\big)(\omega),$$

$$\hat{\varphi}_x(\theta) := \varphi_x\big(u + \theta(v-u), W_u + \theta W_{u,v}, M_u + \theta M_{u,v}\big)(\omega),$$

$$\vdots$$

$$\hat{\varphi}_{txx}(\theta) := \varphi_{txx}\big(u + \theta(v-u), W_u + \theta W_{u,v}, M_u + \theta M_{u,v}\big)(\omega).$$

Henceforth, we implicitly omit $\omega$ to keep the notation lighter. By Taylor's theorem, there exists $\theta \in [0, 1]$ such that $\hat{\varphi}_{xx}(1) - \hat{\varphi}_{xx}(0) = \hat{\varphi}'_{xx}(\theta)$. Hence

$$\begin{aligned}\Gamma^F_{u,v} &= \varphi_{xx}(v, W_v, M_v) - \varphi_{xx}(u, W_u, M_u)\\ &= \hat{\varphi}_{xx}(1) - \hat{\varphi}_{xx}(0)\\ &= \hat{\varphi}'_{xx}(\theta)\\ &= \hat{\varphi}_{txx}(\theta)(v-u) + \hat{\varphi}_{xxx}(\theta)W_{u,v} + \hat{\varphi}_{bxx}(\theta)M_{u,v},\end{aligned}$$

where we have used the chain rule in the last line. For any $p \in (2, 3)$, the function $x \mapsto x^p$ is convex so that we can use Jensen's inequality to see that

$$|\Gamma^F_{u,v}|^p \leq 3^{p-1}C^p\big((v-u)^p + |W_{u,v}|^p + |M_{u,v}|^p\big),$$

where $C = \sup\{\varphi_{txx}, \varphi_{xxx}, \varphi_{bxx}\}$ is just a constant for each path. For almost all paths, $t \mapsto W_t$ and $t \mapsto M_t$ are in $C^{p\text{-var}}$ so that there exists a control function $w(\,\cdot\,,\,\cdot\,)$ with $\|\Gamma^F_{u,v}\|^p \leq w(u, v)^p$. Note that $w(\,\cdot\,,\,\cdot\,)$ depends on each path. Hence $\Gamma^F \in C^{p\text{-var}}$ almost surely. Let us compute the remainder.

$$\begin{aligned}R^F_{u,v} &= \Delta^F_{u,v} - \Gamma^F_u W_{u,v}\\ &= \varphi_x(v, W_v, M_v) - \varphi_x(u, W_u, M_u) - \varphi_{xx}(u, W_u, M_u)W_{u,v}\\ &= \hat{\varphi}_x(1) - \hat{\varphi}_x(0) - \hat{\varphi}_{xx}(0)W_{u,v}.\end{aligned}$$

Using Taylor's theorem, we may now find $\theta \in [0, 1]$ such that $\hat{\varphi}_x(1) - \hat{\varphi}_x(0) = \hat{\varphi}'_x(\theta)$. This yields

$$R^F_{u,v} = \hat{\varphi}'_x(\theta) - \hat{\varphi}_{xx}(0)W_{u,v}.$$

We now apply the chain rule to the definition of $\hat{\varphi}_x(\theta)$ to obtain

$$R^F_{u,v} = \hat{\varphi}_{tx}(\theta)(v-u) + \hat{\varphi}_{bx}(\theta)M_{u,v} + \big(\hat{\varphi}_{xx}(\theta) - \hat{\varphi}_{xx}(0)\big)W_{u,v}.$$

By Taylor's theorem, we can find $\tilde{\theta} \in [0, \theta]$ such that $\hat{\varphi}_{xx}(\theta) - \hat{\varphi}_{xx}(0) = \theta\hat{\varphi}'_{xx}(\tilde{\theta})$. This yields

$$R^F_{u,v} = \hat{\varphi}_{tx}(\theta)(v-u) + \hat{\varphi}_{bx}(\theta)M_{u,v} + \theta\hat{\varphi}'_{xx}(\tilde{\theta})W_{u,v}.$$

Applying the chain rule to the definition of $\hat{\varphi}_{xx}(\tilde{\theta})$ yields

$$\begin{aligned}R^F_{u,v} = \hat{\varphi}_{tx}(\theta)(v-u) + \hat{\varphi}_{bx}(\theta)M_{u,v}\\ + \theta\big(\hat{\varphi}_{txx}(\tilde{\theta})(v-u)W_{u,v} + \hat{\varphi}_{xxx}(\tilde{\theta})(W_{u,v})^2 + \hat{\varphi}_{bxx}(\tilde{\theta})W_{u,v}M_{u,v}\big).\end{aligned}$$

For any $p \in (2, 3)$, the function $x \mapsto x^{\frac{p}{2}}$ is convex so that Jensen's inequality yields

$$\begin{aligned}|R^F_{u,v}|^{\frac{p}{2}} \leq 5^{p-1}\tilde{C}^{\frac{p}{2}}\big((v-u)^{\frac{p}{2}} + |M_{u,v}|^{\frac{p}{2}} + (v-u)^{\frac{p}{2}}|W_{u,v}|^{\frac{p}{2}}\\ + |W_{u,v}|^p + |W_{u,v}|^{\frac{p}{2}}|M_{u,v}|^{\frac{p}{2}}\big),\end{aligned}$$

where $\tilde{C} = \sup\{\varphi_{tx}, \varphi_{bx}, C\}$ is just a constant for each path. By the Cauchy–Schwarz inequality, we have

$$2(v-u)^{\frac{p}{2}}|W_{u,v}|^{\frac{p}{2}} \leq (v-u)^p + |W_{u,v}|^p,$$
$$2|W_{u,v}|^{\frac{p}{2}}|M_{u,v}|^{\frac{p}{2}} \leq |W_{u,v}|^p + |M_{u,v}|^p,$$

so that there exists some control function $\tilde{w}(\cdot,\cdot)$ with $|R^F_{u,v}|^{\frac{p}{2}} \lesssim \tilde{w}(u,v)^{\frac{p}{2}}$. This implies that $R^F \in C^{\frac{p}{2}\text{-var}}$ almost surely. Hence $F$ is $W$-controllable. □

**Lemma A.7** *Consider a derivative whose price at time $t$ depends on the values, also at time $t$, of a set of features given by the $d+2$-dimensional stochastic process $(t, W_t, Y_t^1, Y_t^2, \ldots, Y_t^d)_{t\in[0,T]}$. Then the time-$t$ price of the derivative when $W_t = x$ and $(Y_t^i = y_i)_{i=1,\ldots,d}$ can be expressed as some function $\varphi(t, x, y_1, \ldots, y_d)$. If*

1) *$t \mapsto Y_t^i$ are almost surely in $C^{p\text{-var}}([0,T];\mathbb{R})$ for each $i = 1,\ldots,d$, and*
2) *$\varphi_{xxt}, \varphi_{xxx}, \varphi_{xxy_1}, \ldots \varphi_{xxy_d}$ all exist and are continuous,*

*then the payoff of the option is $W$-controllable.*

The proof is a straightforward adaptation of the proof of Lemma A.6.

***Proof of Theorem 5.19*** For each pair $(t, y)$ in the domain

$$\{(t,y) : t \in [0,T],\, g_1(t) + \epsilon_1 \leq y \leq g_2(t) + \epsilon_2\}$$

and for $i = 1, 2$ and $t \in [0, T]$, we make a smooth choice of diffeomorphism $h_t(y)$ of $\mathbb{R}$ which sends $g_i(t) + \epsilon_i$ to 0 when $i = 1$, to 1 when $i = 2$, and which has derivative 1 in a neighborhood of the barriers.

One can check that $F^{\epsilon_1,\epsilon_2}(g_1, g_2; f)$ has not knocked out by time $T$ if and only if $h_t(W_t) \in (0,1)$ for each $t \in [0,T]$. Thus the payoff of $F^{\epsilon_1,\epsilon_2}(g_1, g_2; f)$ can alternatively be written as

$$F^{\epsilon_1,\epsilon_2}(g_1, g_2; f) = \begin{cases} f(W_T), & \text{if } h_t(W_t) \in (0,1) \text{ for each } t \in [0,T], \\ 0, & \text{otherwise.} \end{cases}$$

Define the process $Y = (Y_t = h_t(W_t))_{t\in[0,T]}$ and denote the running maximum and minimum of $Y$ as $\underline{Y}_t = \inf_{s\leq t} Y_s$ and $\overline{Y}_t = \sup_{s\leq t} Y_s$, respectively, so that $F^{\epsilon_1,\epsilon_2}(g_1, g_2; f)$ can be written as the payoff of an option on the path of $Y$ instead of the path of $W$. We have

$$F^{\epsilon_1,\epsilon_2}(g_1, g_2; f) = (f \circ h_T^{-1})(Y_T)\mathbb{1}_{\{0<\underline{Y}_T, \overline{Y}_T<1\}}.$$

By Itô's formula, we know that $Y$ is a diffusion with smooth coefficients and driven by $W$. Writing $v(t, y)$ for the price at time $t$ of the option with payoff $F^{\epsilon_1,\epsilon_2}(g_1, g_2; f)$ when $Y_t = y$ and assuming the option has not knocked out by that time, we see that $v(t, y)$ satisfies a parabolic PDE with smooth coefficients on

$[0, T) \times (b_1, b_2)$ with boundary conditions

$$v(t, y) = 0, \qquad t \in [0, T), y \in \{0, 1\},$$
$$v(T, y) = (f \circ h_T^{-1})(y), \qquad y \in (0, 1).$$

By Lorenzi and Rhandi [33, Theorem 6.5.2], $v(t, y)$ is smooth in $t$ and $y$ on the rectangle $[0, T] \times [0, 1]$. The solution $v(t, y)$ is also a smooth function of the coefficients of the PDE and of the final boundary condition, which in turn depend smoothly on the choice of the family $(h_t)$ of diffeomorphisms. To see why the solution depends smoothly on the coefficients, we refer to arguments outlined in Henry [27, Chap. 3] which show how to apply the implicit function theorem in Banach spaces together with standard results on existence, uniqueness and well-posedness of parabolic equations to prove such results.

By Lemma A.4, we know that a payoff $G$ is $Y$-controllable if and only if it is $W$-controllable. We use this property to show that the smoothed no-touch payoff is $W$-controllable. The price at time $t$ of an option with payoff $F^{\epsilon_1,\epsilon_2}(g_1, g_2; f)$ is

$$\mathbb{E}_t[F^{\epsilon_1,\epsilon_2}(g_1, g_2; f)] = \mathbb{1}_{\{0<\underline{Y}_t,\overline{Y}_t<1\}} v_h(t, Y_t),$$

where we write $v_h$ to emphasise that the solution of the PDE depends on the choice of diffeomorphism $h$. Hence the price at time $t$ of $G$ when $Y_t = y$, $\underline{Y}_t = \underline{m}$ and $\overline{Y}_t = \overline{m}$ is given by the function

$$\varphi(t, y, \underline{m}, \overline{m}) = \int_{\mathbb{R}}\int_{\mathbb{R}} \frac{1}{b_1}\frac{1}{b_2}\rho\left(\frac{x_1}{b_1}, \frac{x_2}{b_2}\right)\mathbb{1}_{\{x_1<\underline{m},\overline{m}<1+x_2\}} v_{h(x_1,x_2)}(t, y)\, dx_2\, dx_1.$$

In this expression, we write $h(x_1, x_2)$ to emphasise that the choice of diffeomorphism $h$ depends on $x_1$ and $x_2$. Note that it is when writing the expression above that we have used our assumption on the derivative of $h_t$ near the barriers. This ensures that the magnitude of small perturbations of the barrier is unchanged by $h$. It is also at this point that we use the fact that $b_1$ and $b_2$ are assumed to be sufficiently small.

Since $v_y$ depends smoothly on $h$ and $h$ can be taken to vary smoothly with $x_1$ and $x_2$, we may therefore write the price of $G$ as

$$\varphi(t, y, \underline{m}, \overline{m}) = \int_{\mathbb{R}}\int_{\mathbb{R}} \rho(x_1, x_2)\mathbb{1}_{\{x_1<\underline{m},\overline{m}<1+x_2\}} u(t, y, x_1, x_2)\, dx_2\, dx_1$$

for some smooth function $u$. By Lemma A.7, it suffices to show that $\varphi_{yyy}$, $\varphi_{yyt}$, $\varphi_{yy\underline{m}}$ and $\varphi_{yy\overline{m}}$ exist and are continuous. The gamma is given by

$$\varphi_{yy}(t, y, \underline{m}, \overline{m}) = \int_{\mathbb{R}}\int_{\mathbb{R}} \rho(x_1, x_2)\mathbb{1}_{\{x_1<\underline{m},\overline{m}<1+x_2\}} u_{yy}(t, y, x_1, x_2)\, dx_2\, dx_1, \quad \text{(A.5)}$$

and thus the partial derivatives $\varphi_{yyy}$ and $\varphi_{yyt}$ are given by

$$\varphi_{yyy} = \int_{\mathbb{R}}\int_{\mathbb{R}} \rho(x_1, x_2)\mathbb{1}_{\{x_1<\underline{m},\overline{m}<1+x_2\}} u_{yyy}(t, y, x_1, x_2)\, dx_2\, dx_1,$$
$$\varphi_{yyt} = \int_{\mathbb{R}}\int_{\mathbb{R}} \rho(x_1, x_2)\mathbb{1}_{\{x_1<\underline{m},\overline{m}<1+x_2\}} u_{yyt}(t, y, x_1, x_2)\, dx_2\, dx_1,$$

which are continuous by [33, Theorem 6.5.2]. For the remaining partial derivatives, we use that $u$ is smooth in $x_i$, $i = 1, 2$, and the fact that the solution of the PDE depends smoothly on its coefficients as discussed above. Using integration by parts shows that

$$\begin{aligned}\varphi_{yy\underline{m}}(t, x, \underline{m}, \overline{m}) &= \int_{\mathbb{R}} \mathbb{1}_{\{\overline{m}<x_2\}} \int_{\mathbb{R}} \mathbb{1}_{\{x_1<\underline{m}\}} \frac{\partial(\rho(x_1, x_2)u_{yy}(t, y, x_1, x_2))}{\partial x_1}\, dx_1\, dx_2 \\ &\quad - \int_{\mathbb{R}} \mathbb{1}_{\{\overline{m}<x_2\}} \big(\mathbb{1}_{\{x_1<\underline{m}\}}\rho(x_1, x_2)u_{yy}(t, y, x_1, x_2)\big)\big|_{x_1=-\infty}^{x_1=+\infty}\, dx_2 \\ &= \int_{\mathbb{R}} \mathbb{1}_{\{\overline{m}<x_2\}} \int_{\mathbb{R}} \mathbb{1}_{\{x_1<\underline{m}\}} \frac{\partial(\rho(x_1, x_2)u_{yy}(t, y, x_1, x_2))}{\partial x_1}\, dx_1\, dx_2,\end{aligned}$$

where the last term in the penultimate line vanishes because $\rho$ has compact support. As $\rho$ and $u_{yy}$ are smooth in their parameters, we deduce that $\varphi_{yy\underline{m}}$ is continuous. A similar calculation shows that $\varphi_{yy\overline{m}}$ exists and is continuous, as required.

The controllability of the payoff follows by applying Lemma A.7. To show that it is continuously controllable, we note that

$$\|(S, M) - (\hat{S}, \hat{M})\|_{p\text{-var};[0,T]} \lesssim \|S - \hat{S}\|_{p\text{-var};[0,T]}$$

since the paths of $t \mapsto M_t$ are at least as regular as those of $t \mapsto S_t$. It is now straightforward to obtain Lipschitz bounds on the $\Delta$, $\Gamma$ and remainder terms by using formulae of the type obtained in Lemma A.7. □

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## Declarations

**Competing interests** The authors declare no competing interests.

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